Gray Areas in College Athletes' Data and Privacy

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ACM Reference Format:
Primary Author and Secondary Author. 2026. Gray Areas in College Athletes' Data and Privacy. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (ACM Conference 2027).* ACM, New York, NY, USA, 61 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

College athletes have reached an elite level of sports competition. They have exceeded many of their pre-college peers and some of them aspire to continue progressing to still higher levels in their sport. College sports teams regularly gather health and performance data on athletes. This sociotechnical system has been termed "sports science" and functions to improve athletic performance and to reduce athletes' risk of injury. Some college teams, often those with larger budgets and who play at higher levels of inter-collegiate competition, invest greater resources in technology to collect athletes' data. Personal wearable devices have become prevalent enough that some college athletes augment their teams' data collection effort with their own data, whether independently or in collaboration with coaches or teammates.

The use of data by college sports has grown significantly in recent years[ref] as teams seek competitive advantage and a better, healthier experience for college athletes who now have more freedom to change schools, under changing eligibility rules dictated by U.S. inter-collegiate sports' governing body, the NCAA. The changes in college sports with regard to player movement have increased the need for financial resources and the demand for data resources by university athletic departments[ref].

Matthew Weiss comes to trial in June, 2026, in federal court for illegally accessing thousands of college athlete records by breaching the security of software used by athletic trainers. Weiss has also been sued in a civil class-action lawsuit that involves 15,000 victims[ref]. Data security and privacy is not a subject that not many student-athletes receive information about as they devote considerable time and energy to their athletic and academic pursuits. As the network of

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ACM Conference 2027, Location TBD

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/XXXXXXX.XXXXXXX

interconnections among users of athletes' health and performance data grows, so too does the risk and challenge to maintain personal privacy.

As a starting point that we hope will lead to a broader discussion of student-athletes' health and performance data privacy, we ask these research questions:

- RQ1: What are athletes' privacy expectations for ongoing tracking of their health and performance data by college sports teams?
- RQ2: What are athletes' privacy expectations for data sharing when it comes to their health and performance data?
- RQ3: What are athletes' privacy expectations for downstream use of their health and performance data?

To address these questions we apply reflexive thematic analysis to interview transcripts from 16 inter-collegiate student-athletes. We also conducted surveys prior to our interviews. The interviews and the preliminary surveys are based on the contextual integrity (CI) framework[ref]. CI is a useful method for understanding privacy expectations by describing the context of how information flows among its users for a specific purpose. Student-athletes play a central role in the information flows for health and performance data in the college sports context. We describe a vivid picture of how athletes' individual development and their teams' collaborative efforts proceed and what student-athletes understand about their data privacy in this context.

Our research questions set out to determine what are college athletes' privacy expectations. We used the CI framework to guide the content and structure of the questions in our interview study The CI framework also informed the reflexive thematic analysis and the data visualizations of information flow diagrams. Because student-athlete participant responses are highly variable, findings related to our research question have been supported with additional quantitative evidence that we collected from interview participants, and from external surveys of similar athlete populations.

The work contributes a fundamental understanding of student-athletes' privacy expectations given the range of health and performance data types that teams collect, and which student-athletes sometimes collect for themselves. We found that tension exists between an athletes' personal student-athlete identity, which we associate with athlete-centric information flows, and the social identity that occurs on a sports team, which we associate with team-centric information flows. Individual student-athletes differ widely in their data privacy expectations but our findings demonstrate that the resolution of what they believe is good for themselves as an athlete and what they believe is good for their team drives student-athlete expectations for privacy.

2 Related Work

This study relies on the contextual integrity (CI) framework for the design of the surveys and interview script we used with our student-athlete participants. Other related work draws on human-computer interaction research that explores college athletes, their data, and technologies used to collect, manage, and report their information. The sports psychology literature is a helpful source for understanding student-athletes' self-identity and the social identity that comes with playing on a team.

2.1 Contextual Integrity

Contextual integrity is useful for evaluating the privacy attitudes and behaviors of target population[ref]. We apply the framework to student-athletes in the college sports context. We developed an interview script for semi-structured interviews with student-athletes that reflects the CI framework and we employed surveys, use cases and hypothetical scenarios to elicit responses.

The contextual integrity framework on which this study is based on investigates student-athletes' sport-related health and performance information flows and seeks to determine whether an information flow is appropriate. CI asks what are the privacy norms of a situation, and based on those norms, are the student-athletes' privacy expectations met or unmet[ref]. When privacy expectations are met, and norms for shared information are respected, then personal information flows should be considered appropriate.[ref]

The CI framework has a set of five parameters that describe a contextual norm related to personal information flows. A *data type* describes the shared information. A *subject* describes who the information is about, usually it is the source of collected data. A *sender* describes who, or in some automated instances what, shares the data. A *recipient* gets the data. A *transmission principle* imposes constraints on the flow of information. An additional sixth parameter, purpose, is relevant to this study. *Purpose* describes the reason for having and sharing personal information, which is essential to the context.[ref]

2.2 Other Related Work

This study relies on the contextual integrity framework for the design of the surveys and interview script we used with our student-athlete participants. Other related work draws on human-computer interaction research that explores college athletes, their data, and technologies used to collect, manage, and report their information. Related work in sports psychology informs the coexistence of athletes' individual autonomy and their socially-informed collaboration with coaches, team staff, and teammates.

The coach-athlete relationship dyad is a high-stakes situation where authority lies with the coach, and where that authority may or may not adversely affect the autonomy of a student-athlete. Kolovson et al. used boundary negotiation artifacts, essentially reports based on student-athletes health and performance data, to illustrate the power asymmetry that exists in college sports' coach-athlete collaboration[ref]. Later work by Kolovson et al. showed how these asymmetries between coaches and athletes do not always lead to outcomes in the athletes' best interests.

The data used in the process of athlete development is U.S. college sports has led to compelling work by Brewer, Childs et al. in the field of Sports Human-Computer Interaction. Brewer et al. has documented coaches data practices, concluding that these practices seek to balance performance goals, athlete emotional well-being, and privacy[ref]. The subsequent investigation by Brewer et al. into how student-athletes engage with sports data showed a fundamental tradeoff between athletic development and well-being[ref]. Additional work by Childs, Brewer et al. documents CI information flows and outlines privacy protections for college athletes who use continuous wearable devices[ref].

The work by Clegg et al. into student-athletes' learning with sports data is a helpful guide to their hands-on experience with data collected in the college sports context[ref]. Further work by Clegg et al documents information asymmetries, where athletes see little of the data collected on them, and where these athletes are uncomfortable with the sharing of their data that goes on[ref].

Organizations that oversee sports confederations and leagues will conduct surveys of participating athletes, and sometimes publish them. The NCAA oversees college sports in the U.S. and conducts regular, voluntary surveys of student-athletes[ref]. Survey data that touch on athletes' data collection, management, and reporting that comes from the NCAA and other sports oversight organizations are useful guideposts and checks related to our research questions on student-athletes' privacy expectations.

Sports psychology refers to athletes' individual personal identity as their athlete identity[ref]. College athletes will also have a distinct student identity that can complement or challenge their athlete identity[ref]. We refer to our interview subjects as student-athletes as a reminder that our questions target both the student and the athlete identities of the people we spoke with.

Sports psychology research has identified social identity as the functional term to describe how athletes navigate their personal identities and the group identity that comes with belonging to a team[ref]. Social identity theory attempts to understand inter-group relations and predates its application to sports contexts[ref]. The approach offers a way to understand how individual psychology both structures, and is structured by, the dynamics of group life[ref]. The boundaries between personal and group identities are fuzzy. We use the term "athlete-centric" to describe information flows that map to student-athletes' personal identity and the term "team-centric" to map to student-athletes' social identity.

3 Methods

The qualitative analysis of interview transcripts was helped by quantitative word frequency analysis. The qualitative analysis also relied on reflexive thematic analysis to hone a detailed understanding of information flows formed by application of the CI framework. We checked our qualitative understanding of student-athletes' attitudes with visual diagrams to show the flows of information, with quantitative analysis of our transcripts and surveys, and with comparisons to surveys of similar college and professional athlete groups.

3.1 Study Design

We conducted face-to-face semi-structured interviews with 16 college student-athletes who play NCAA Division I intercollegiate sports. Two surveys preceded the actual interview. One survey was in place to screen participants in the study and collect basic information (e.g., gender, age, sport).

We gave a second survey, a Personal Tracking Technology Questionnaire (PTTQ), at the start of our in-person sessions in order to gather the range of data types that the student-athletes experienced in their college sports participation. The comprehensive list of data types (i.e., the first parameter of the CI framework) includes 4-7 examples in the following categories: Fitness Testing, Training Load, Skill Development, Movement/Symmetry, Body Composition, Strength Training, Jump Training, Head Impacts, Speed Training, Endurance Training, Sleep Recovery, Non-sleep Recovery, Nutrition/Metabolism, Fluid Balance/Hydration.

Interviews started with a set of foundational questions designed to put student-athletes' relationships with peer students, teammates, and coaches in the context of a biopsychosocial development paradigm where physical health, mental well-being, and interpersonal relationships all contribute to athletic success.

3.2 Applying Contextual Integrity

The rest of the interview consisted of use cases, each with a related hypothetical scenario. The six use cases that map CI parameters were Training Load, Sleep and Recovery, Injury, Setbacks, Nutrition, and Womens[sic]. The Womens use case was exclusively for women athletes and focused on data related to menstrual tracking. A seventh use case, Leaderboards, is an example of an end-to-end CI information flow.

Interview questions and hypotheticals in the seven use cases were meant to capture student-athlete attitudes and behaviors and to identify CI parameters for the information flows based on the different use case contexts.

The Training Load use case is a representative example of a CI data type used by college coaches and athletes. Training load in practice attempts to measure the total amount of work that athletes are doing during training, practice and competition with objective (like running speeds and distances) and biometric data (like heartrate or body composition).

The Sleep and Recovery and Nutrition use cases highlight how athletes' and collaborators navigate performance aspects outside of the physical work of training. Injury and Setbacks use cases are complementary. Injury focuses on injury prevention and occurrence and Setbacks focuses on athletes dealing with a lack of progress in their development (whether from injury or another cause).

The Leaderboards use case presents a common, sometimes problematic[ref], example of an information flow used on college sports teams. The Womens case study looks at menstrual cycle tracking, something that highlights an emerging research area and also considers gender differences in athlete information flows.

The student-athlete in a college sports context experiences data types collected for their personal athletic development, and for their collaborations with teammates and coaches. These are two distinct socio-normative strata and they are the primary contexts for athletes' data collection, analysis, and sharing. We identify the personal athletic development context associated with athlete identity as an *athlete-centric* information flow and we identify the athletes' collaboration with their team as a *team-centric* information flow associated with the social identity that comes with team participation.

Another simplifying assumption that we make is to say that an individual athlete is always the subject, the initial CI parameter at the start of an information flow. Senders and receivers, which in CI terms are the intermediary and destination in an information flow, can vary among the set of teammates, coaches and athlete support personnel that constitute the student-athletes' team. Information flows with receivers who are outside the team can also exist in secondary contexts, something we also consider.

3.3 Word Frequency Analysis

Word frequency analysis provided a useful starting place for analyzing interview transcripts, providing needed assistance in differentiating athlete-centric and team-centric information flows.

An interesting linguistic pattern exists for athletes who play on teams, and all of our student-athlete participants play on teams. We counted instances where the student-athlete described personal attitudes and behaviors with "I" first-person singular subject pronouns. We also counted instances where the student-athlete referred to their team with "we" first-person plural subject pronouns. Microsoft Excel has a keyword counting function that we used to count personal pronouns in the interview transcripts.

We recognize the difference between participants' "I" personal pronoun responses and their "we" personal pronoun responses are indicators of either self or team perspectives in answers to interview questions. Responses with the word "coach" were also indicators of team, as opposed to personal, references. The word "coach" was also counted. The count for I-responses divided by the total count for we- and coach-responses' counts created a ratio that showed participants' personal versus team perspective for each of our study's different use cases.

The calculated I-We ratio does not have a precedent in contextual integrity research. Personal pronoun counts in other research fields considered them in relation to total words[ref]. The singular college sports context which involves the individual athlete-centric context and the collaborative team-centric context makes this type of word frequency analysis practical and useful.

The I-We ratio was at first a tool for initial exploratory data analysis. It was easy to see differences between and patterns among participants' I-We ratios using radar plots drawn in the Observable data visualization programming language. Those radar plots (Appendix B) would, as the analysis became more refined, get reformatted into the results Table 1 that shows the I-We ratios for participants and use cases.

3.4 Qualitative Analysis

Our qualitative analysis used reflexive thematic analysis (RTA), as outlined by Braun and Clarke[ref]. This thematic analysis method is appropriate for one-person coding and it supports inductive and deductive coding practices. We surfaced inductive codes and themes based on the text, and we also introduced CI-oriented codes and themes from outside the text.

RTA is a dynamic process. Each of our five passes of codes focused on narrower and narrower aspects of the CI framework, working toward the presentation of a complete picture of how contextual integrity fits in the college sports context.

3.4.1 Reflexive Statement. Compared to other thematic coding methods, RTA is closer to the primary coders' experience as a professional journalist and undergraduate English major familiar with evidence-based reporting and critical theory in literature. The primary coder was also an inter-collegiate basketball player as an undergraduate and who, in recent years, consulted for professional sports organizations (U.S. Soccer Federation, Denver Broncos, Chicago Cubs, Arsenal Football Club) and has worked on commercial athlete management system software.

3.4.2 Reflexive Thematic Analysis. Reflexive thematic analysis progressed through five distinct passes through our interview data, starting with initial codes that distinguish essential aspects of individual athlete development and of athletes' collaborations for team-related purposes. The first pass used straightforward fundamental codes to shed light on the most basic aspects of what is personal athlete development and what is team-related activity. The subsequent passes compound on the first set of codes and themes, building toward a complete representation of our interviews' text.

3.4.3 RTA Version 1. The work that athletes do in training to improve strength and fitness makes them more athletic for their sport and less likely to experience injury, but only if the athletes do not overextend. We used codes that match the three pillars of the athletic biopsychosocial development model, plus a fourth code to identify challenges to develop the "Athlete Development" theme.

College sports teams collect data to monitor student-athletes workloads during training, practices, and games. We used codes to document the primary athlete-collaborator combinations (head coach, assistant coach, strength coach, trainer, teammates) in order to develop the "Team Collaboration" theme.

3.4.4 RTA Versions 2-3: Inductive. The second RTA pass sought to expose whatever tension or conflict might exist in college sports information flows, details that help us to understand where, when and how inappropriate information flows occur. A few student-athletes that we spoke with identified tension between athlete-centric and team-centric information flows and called it a "gray area." Codes for conflicts and for athletes' personal choices that could lead to conflicts were developed into the "Gray Areas" theme.

An insight that cut across three different use cases led to our third pass RTA. We asked student-athletes about injury risk and injury prevention in the Injury use case. Among the student-athletes we spoke to, injury prevention was an expression of their positive habits, and injury risk was whether or not they followed through with those positive habits. We also asked our student-athletes about their habits, specifically related to sleep and nutrition, in two other use cases.

College athletes are aware of how important habits are for their own and their teams' success, but these are young people, and habit formation is an active learning process for them. "Injury risk is, depending on like how you're living, like like your sleep and nutrition," one athlete said, adding that "it can put you out like a higher chance of like having a certain injury if you're not getting enough sleep, not nutrition, nutrition is strong as well."

The scope of our codes for this part of RTA expanded from just habits to a range of behaviors like role models, information seeking, improvement strategies, and external resources. Those codes were developed into the "Set up for Success" theme. How student-athletes set themselves up for success is integral to how they contextualize and differentiate athlete-centric and team-centric information flows in the process of their own biopsychosocial development.

3.4.5 RTA Version 4: Deductive. The Leaderboard use case is a self-contained information flow in addition to being common practice for college sports teams. We wanted to see what we could learn from this micro instance and build on what we learned with our first pass themes for Athlete Development and Team Collaboration. Our codes sought to capture the differences among athletes in their perspectives on leaderboards, and the different rationales for coaches instituting leaderboards. "Differentiation" and "Presentation" are the two themes we developed and they are more in-depth investigations into the Purpose and Benefit themes of our second pass RTA.

3.4.6 RTA Version 5. The fifth and last pass was an effort to tie up loose ends and ask what's missing with regard to the CI framework. Athletes told us that they defer privacy decisions to professional collaborators (coaches, trainers) but not to teammates, so we coded for important aspects of these jobs with regard to privacy decision-making and developed the "Role and Responsibility" theme. We also noticed that the word "friend" occurs once in the seven use cases for all of our interview transcripts, something that made us want to code for the relationship dynamics at work among student-athletes and collaborators and develop the "Relationship Dynamics" theme. This information, we feel, gives us a better handle on senders and on transmission principles between senders and athlete-recipients, two essential CI parameters.

3.5 Flow Diagrams

We surveyed student-athlete subjects before starting our interviews to gather information for CI data types and CI recipients regarding the CI data types collected related to their college sport participation. We rely on the assumption the student-athlete themselves was always the CI subject, the other two CI parameters (sender and transmission principle) could be deduced from our interview responses.

The CI parameter information was formatted in json (data-interchange format) so that we could visualize representative CI information flows for the range of CI data types that each student-athlete in our study experiences. We use Sankey diagrams created in Python using Plotly to show the information flow as a directed graph showing what occurs as a data type progresses through the CI parameters in an information flow.

These diagrams are important for making sense of and seeing the logic in CI information flows. If the information flow is clear and understandable then we probably have a cogent, explainable representation for what is going on. If the information flow is spaghetti then we have more work to do in order to understand and explain what is going on.

3.6 Participants

Our study recruited student-athlete study participants using flyers and with direct outreach to the university Department of Exercise Science and to the athletic department's Student-athlete Association. Snowball sampling was also used. Study design and recruitment were approved by the our university's Institutional Review Board, as was the payment of $30-40 gift cards to study participants. NCAA rules for student-athletes prohibit payments that exceed research payments to the general student population. Twenty-seven athletes completed initial surveys and 16 students completed in-person interviews (12 women and 4 men). Athletes participate in 7 different inter-collegiate sports (Cross Country, Skiing, Track, Swimming, Field Hockey, Soccer, Lacrosse) and play on 11 different mens or womens inter-collegiate sport teams.

4 Results

After a handful of student-athlete interviews we realized that college athlete opinions and behaviors are highly variable. Our initial study design had an assumption that we should recruit for gender and sport diversity in order to have a range of perspectives that could be representative of the larger, general college athlete population. We also realized that the range of student-athletes' opinions varies even when athletes have the same gender or play the same sport, meaning that our diversity recruitment was not as important as we first thought. The individual athletes we spoke with, when asked what they think, gave us very different opinions.

4.1 RQ1: Ongoing Tracking

The overwhelming majority of college athletes' health and performance data is in the present and very near-future. Training and training-related activities are, in nearly all cases, in the present tense and take place in preparation for the next competitive event.

Table 1. I/We Ratio for use Cases

	Use Cases						
Participant	Training Load	Sleep	Injury	Setbacks	Leaderboard	Nutrition	Womens
1	1.0	1.8	2.8	2.1	5.3	2.9	23.0
2	1.3	6.0	3.4	12.0	2.3	7.0	30.0
3	0.6	3.9	2.0	4.2	0.9	2.5	-
4	1.1	6.9	2.4	1.9	3.5	4.9	26.0
5	4.5	3.8	1.1	2.7	2.0	2.9	2.3
6	0.8	1.4	1.2	2.4	2.7	4.3	4.3
7	0.8	2.7	46.0	5.0	3.5	5.5	15
8	1.6	5.8	4.7	7.3	2.5	3.7	11.0
9	3.5	2.8	3.4	7.8	0.8	2.9	16.5
10	1.7	10.0	2.7	15.5	4.0	2.4	3.2
11	2.5	21.0	3.3	5.0	9.5	3.3	-
12	0.7	11.0	2.8	2.4	0.0	15.0	2.2
13	1.8	9.0	2.9	3.5	6.0	8.0	14.0
14	1.3	9.0	3.0	4.0	3.0	1.9	18.0
15	0.7	2.9	3.2	2.8	2.0	1.9	-
16	2.8	24.0	2.5	31	10	3	-
mean	1.7	7.6	5.5	6.8	3.6	4.5	10.3
std	1.1	6.6	10.8	7.5	2.8	3.3	10.2

I/We ratio is the frequency the pronoun "I" in a use case divided by the frequency of the "we" pronoun plus the frequency of the word "coach" in the same use case.

4.1.1 *I/We Ratio.*

Word frequency analysis gave us a way to understand just how different the athlete perspectives were and to clearly differentiate between athlete-centric and team-centric information flows for student-athletes' data. The I/We ratio calculations for all seven use cases are shown in 1. Large values appear intermittently, and these spikes are indicative of the student-athletes' personal involvement, firm beliefs, or strong opinions related to the issues presented in a particular use case. There is no real trend or pattern except to say that some athletes feel strongly about some things.

Patterns that did emerge are subtle. For example, the "Training Load" use case has lower mean and standard deviation values compared to the other six use cases. Based on the rationale for the I/We ratio, training load data collection, sharing, and use has more of a team-based collaboration context than an individual athletic development context compared to the data described by the other six use cases.

4.1.2 *Flow Diagrams in Practice.*

Initial flow diagrams showed all of the athletes' data types in a single visualization (examples in Appendix C). These diagrams were far too convoluted to generate any meaningful insight. The nodes and links in a comprehensive representation of athletes' CI information flows produced a spaghetti of lines and blobs that had little utility.

Once we started to ask more specific questions of our CI parameter data we started to see how the data collected on athletes was used. We separated the information flows for athletes' personal health and performance data, typically gathered by using a personal continuous wearable biometric device, from the data collected by their teams during training, practices, or competitions. *Differences between athlete-centric and team-centric information flows were apparent.*

4.1.3 Thematic Analysis. In the "Gray Areas" theme we report what a student-athlete said in response to the question, "What do you mean by gray area?" "It's the things that kind of affect training and it could be technically part of the team's business because of that," the athlete told us, "but also, it's kind of like part of your personal life." The gray area is where data used by the team encroaches on the athlete as a person.

The "Set up for success" theme leaned into habits, role models, information seeking, and external resources as data-centric paths to athletic and personal development. "The sleep habits that made my sleep better, a big thing was sleep consistency." "Everyone use to be like super skinny" and "now everyone's kind of like embraced getting stronger." "I actually did a really interesting study about this of how estrogen levels correlate with ACL tears." "I would take my Whoop and tape it so I could have that data. And sometimes like if I had a bad game or something, I would still look at the data." These are representative quotes describing ongoing interaction the student-athletes' data.

4.1.4 Representative Use Case: Training Load. The low values for participants' I/We ratio in the Training Load use case occurred even though five participants reported that their team did not collect training load data. Two of those five athletes did gather and use training load data on their own, using a personal continuous wearable device. Even though Training Load has more of a team orientation than other use cases, individual athletes will still look to this data for their personal athletic development with an athlete-centric information flow.

Athletes and coaches refer to the term "training load" as a comprehensive measurement of the work undertaken by individual athletes during training, practicing, and competing in their sports. The goal of monitoring training load is to avoid overuse injuries while maximizing the time and effort devoted to the teams' physical preparation, skill development, and team cohesion[ref].

Information flow diagrams show the path of athletes' data through the series of contextual integrity parameters. The information flows for athletes' training load data in a team context are shown in Figure 1. CI data types and CI recipients are data collected in surveys. The other CI parameters have been fleshed out using interview responses. Seven of our interviewees have a training load information flow where coaches, usually strength coaches, act as a hub to gather and disseminate information from athletes' data. These information flows show the extent of the collaboration using athletes' data between the athletes and coaches.

Thematic analysis supports centrality of training load for understanding what's going on with athletes' data. The first RTA pass covered fundamental aspects of athlete development, intra-team collaboration, and information flow. The Training Load case was emblematic in each regard.

"The individual athlete benefits 'cause it gives them a better idea of their own training. I think the coaches can benefit because certain coaches in certain instances, they can use that to better understand their athlete and what's best for that athlete," said one student-athlete about training load data, as collected in our Information Flow theme.

"It's super easy to overextend. You might even notice that you might be feeling OK, but overdo it and overtraining is a real thing," reflect the physical aspects of the biopsychosocial model for

Fig. 1. Training Load Use Case Information Flows

athlete development. "I feel like there's a difference between mental and physical readiness. What you need to do mentally to get ready makes you feel like you're physically better," speaks to the mental aspects of the biopsychosocial model.

Related to the Team Collaboration RTA codes, one athlete said that "the assistant coaches generally don't make the training plan, but he does. He will look at the difference between the training what I was supposed to do and the one I was gonna do" in order to correct the situation.

Questions about the purpose for and who benefits from athletes' data are pertinent to CI information flows. One student-athlete said during the Training Load use case, "I think the purpose is for our coaches and our team to stay organized and make sure no one's doing too much or too little for themselves." Another said, "The individual athlete benefits 'cause it gives them a better idea of their own training. I think the coaches can benefit 'cause it. I mean, certain coaches in certain instances they can use that to better understand their athlete and what's best for that athlete."

4.2 RQ2: Data Sharing

Data sharing has two types in the college sports context, inside and outside of the team. The Leaderboard use case is a good example of sharing inside the team. We used a questionnaire to ask student-athletes about the different possible CI recipients (themselves, Head coach, Trainer, Assistant coach, Teammates) their data might wind up with. There two hypothetical scenarios that

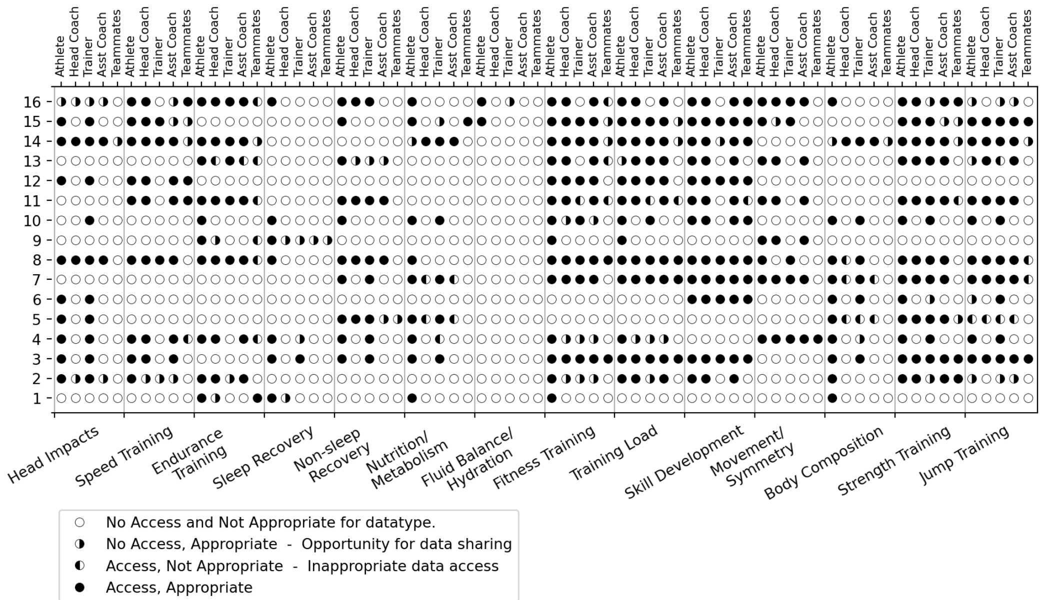


Fig. 2. Personal Tracking Technologies Questionnaire Responses

asked student-athletes to consider sharing their data with outside researchers in an effort to probe sharing outside the team.

4.2.1 Data Sharing Inside of the Team - Questionnaire Responses. We summarize student-athlete responses to a questionnaire that we administered prior to the interviews in Figure 2 and Table 2. We asked the student-athlete who among themselves, the Head Coach, Assistant Coaches, Trainers, and Teammates have access to 14 different CI data types, and who among the same groups would be appropriate CI recipients for the data types.

The data types with the most sharing (full circles and right-black circles) are the team training and practice activities (Skill, Strength, Fitness, Training, Jump). The data types with the least sharing are more personal activities (Sleep, Nutrition, Body Composition).

Nutrition, Body Composition, and Endurance show troublesome percentages of inappropriate sharing (left-black circles), all of them greater than 20%.

4.2.2 Data Sharing Inside of the Team - Hypothetical Scenarios. Each use case in the interview includes a hypothetical scenario. The hypothetical for the Training Load use case asked whether it was acceptable to share training load data with researchers working to create algorithms that predict future athletic performance. We asked athletes how they felt if their teams' required them to follow algorithmic training recommendations, and if their teams shared their data with researchers without explicit approval. Athletes expressed high levels of support for their team in response to both questions, as shown in Table 3a.

The Setback use case defines a setback as "a time when you were not able to reach your training goals," and we have a use case asking interview subjects questions about setbacks they have experienced. The use case discussing Injuries precedes the Setbacks case. To avoid overlap with the Injuries case, Injuries covers the experience of being injured where Setbacks has multiple questions about overcoming the student-athlete's "setback" that is unspecified. The Setbacks use case has

Table 3. Hypothetical Scenarios: Appropriateness and Data Sharing

(a) Training Load			(b) Setbacks			(c) Leaderboard		
Researchers develop algorithms that predict performance based on training load data. The researchers make it available to your team.			**College teams use athletes' performance tracking technology data to manage athlete development setbacks.**			**Coach and trainer each have a leaderboard. The trainer says that the leaderboard describes speed in a singular context and the goal is to improve players' performance. The coach wants to spark competition throughout the team and says that the athletes at the bottom are "holding the average down."**		
Would you follow the algorithms' recommendation?	How would you feel if your team required you to follow the recommendation?	Is it acceptable for your team to share your training load data with the researchers?	How do you feel about it?	Are there specific purposes or contexts that make it appropriate?	A coach has a secret KPI for evaluations and decides that your role is going to be reduced based on it. What would you do?	College athletic teams' leaderboards: appropriate?	Single context training leaderboards: appropriate?	Competition throughout the team leaderboards: appropriate?
◐	●	◐	○	◐	◐	◐	●	◐
●	◐	●	●	○	○	●	●	○
●	●	◐	●	◐	◐	●	●	◐
●	●	◐	●	○	○	●	●	◐
◐	●	◐	●	◐	○	◐	●	○
◐	●	●	●	◐	◐	○	◐	○
●	●	◐	●	◐	◐	○	◐	○
●	●	◐	●	◐	◐	◐	◐	◐
◐	◐	●	◐	○	◐	●	●	○
●	○	◐	●	◐	◐	○	◐	○
◐	◐	◐	●	◐	○	◐	●	◐
○	●	●	◐	◐	●	○	◐	○
●	●	○	◐	○	◐	◐	◐	○
◐	◐	○	●	●	◐	●	●	○
●	●	●	●	●	◐	●	●	●
●	◐	◐	◐	◐	●	○	○	○

Key: ● full support ◐ some support ○ no support

consistently higher I/We ratio numbers than than the Training Load use case and the injuries use case, as shown in Table 1.

We asked student-athletes how they felt about college teams using performance tracking technology to manage athlete development setbacks, and whether there were specific purposes or contexts that would make the use case appropriate (or inappropriate). Responses showed less support for the team than with the Training Load use case, as shown in 3a.

We used a hypothetical scenario to ask student-athletes to compare two different types of leaderboard situations and tell use whether the situations were appropriate. In one situation a hypothetical trainer is careful to point out that the leaderboard describes speed in a singular context, and the goal is to improve players' performance in those situations. The other situation has a coach who wants to spark competition throughout the team, and points out that the athletes at the bottom are "holding the average down."

Our interviews subjects said the leaderboard that had the specific purpose was more appropriate than the leaderboard designed to spark competition throughout the team, showin in Table 4.

Table 2. PTTQ Summary: Access and Appropriate Sharing

Data type	**Athlete-centric only**	**Team-centric only**	**Athlete- & Team-centric**	**Inappropriate to Share**	**More Sharing: Athlete-centric**	**More Sharing: Team-centric**	**More Sharing: Both**
Head Impacts	0	1	9	0	0	2	1
Speed Training	1	0	8	1	0	4	0
Endurance Training	3	0	7	6	0	2	0
Sleep Recovery	6	0	1	0	0	3	0
Non-sleep Recovery	3	0	7	6	0	2	0
Nutrition/Metabolism	4	1	5	3	1	1	0
Fluid Balance/Hydration	2	0	0	0	0	1	0
Fitness Testing	4	0	10	3	0	5	0
Training Load	2	0	6	1	1	4	0
Skill Development	0	0	12	1	0	1	0
Movement/Symmetry	0	0	8	0	0	1	0
Body Composition	4	1	6	3	0	1	1
Strength Training	1	0	12	1	0	6	0
Jump Training	0	2	9	4	2	0	2

"Athlete-centric" frequency is where the student-athlete does not select a coach or trainer for a datatype. "Team-centric" frequency is where the student-athlete does not select the athlete themselves or teammates. "Athlete- & Team-centric" frequency has both types of responses. "Inappropriate to Share" is the frequency that a datatype is accessed, but the access is not considered appropriate. "More Sharing" frequencies reflect where student-athletes believe that the athlete, or a coach or trainer, or teammates should, but do not have access to a datatype. All student-athletes did not use all datatypes.

4.2.3 Data Sharing Inside of the Team - Reflexive Thematic Analysis. An athlete described their team's leaderboard in the RTA first-pass Collaboration theme, saying it's "not explicitly who's doing well and who's not doing well like that black and white. It's more of like a spreadsheet of data on where we are for our times and distances. And that's available to everyone." By looking more closely in this pass we were able to be more specific. "I guess the fastest people benefit. Yeah. I mean all of us benefit though, because we all want the fastest team to go," one student-athlete said, something that coded for "Mudita" in our Differentiation theme. Mudita refers to a Buddhist expression that means feeling happiness and fulfillment in the success of others.

Coaches like leaderboards. They check a lot of boxes for motivation, education, and accountability, which is reflected in the Differentiation theme. "If I'm doing well and I'm like next to them or even beating it, beating them, that's when I know I'm doing good," said a student-athlete. "I think everyone sees someone else's numbers and then starts talking and starts trying to prove themselves. I think it promotes a healthy environment," said another student-athlete.

Leaderboards are sometimes created without explicit intent, almost by accident, because it can be convenient to collect athletes' data when they are together as a group. "There's a sheet with all of our names and we write down what time we got for that rep, and you can do like a visual scan that gets create like kind of like a mental leaderboard," according to one athlete.

The data sharing on a leaderboard does not benefit all of the athletes on a team equally. Athletes at the top benefit, more than the athletes below them. "You only compare yourself and instead of wanting to be better, you're just thinking you are less because you are less in paper," said one student-athlete, included in our "Differentiation" theme.

The "Role and Responsibility" theme in our RTA showed that they think more in terms of roles that coaches perform. Two roles, decision-maker and planner, seem to factor most heavily in what our interview subjects see coaches doing with athletes' data. There are also expert support roles for strength coaches, athletic trainers, and nutritionists who have know how for collecting and interpreting data, and that know how is important for athlete development. Head coaches, assistant coaches, and strength coaches have more team-centric information flows, as shown in the Training Load use case Figure 1. Athletic trainers and nutritionists deal with individual athletes, and have more athletic-centric information flows.

Those roles (decision-maker, planner, expert) have responsibility for teams' data. Student-athletes we spoke to acknowledge the control over players' data that those roles exercise. When we asked about data sharing with researchers in hypothetical scenarios, student-athletes felt it was mostly within the scope of their roles' responsibilities.

The "Relationship dynamics" theme in our last-pass RTA gives voice to how data fits into athlete-team collaborative interactions are like. Sub-themes of progression ("I love doing that with my coach to talk and like get a gauge of like where she thinks my progress is."), adaption ("Say we had like a crazy hard practice and our loads reflected that, or like a game, reflect that in the lift that we do."), consensus ("If that was a team decision and everyone was moving forward with it, then I would buy into that."), and maturity ("The next summer while I trained, I used my major and the knowledge I gained to train more efficiently.") reflect different drivers for the relationship dynamics.

4.2.4 Not Sharing Data Inside of the Team. A substantial percentage of student-athletes we spoke would like more data collected and shared with them than teams currently collect and share. 4/15 responses said that the their teams collect "too little" data and 6/14 said that their teams share "too little" data with them. The desire for more information shows the propensity for information-seeking among young athletes.

PTTQ responses in Table 2 show a stronger wish for more data sharing in the team-centric context than in the athlete-centric contexts. The CI data types that represent team activities (strength, fitness, training) are both athlete- and team-centric information flows, but these activities have the most opportunity for still more data sharing among team coaches, trainers, and teammates.

We made sure to list exceptions in our "Role and Responsibility" theme for when the coach's or trainer's role would not necessarily be appropriate for access to a datatype. For example, as student-athletes told us, "A coach isn't going to benefit too too much directly from like knowing that I got 8 hours," and "I usually don't like sharing my information to my male athletic trainers because they wouldn't be able to understand what I mean."

4.2.5 Data Sharing Outside of the Team. A large majority of student-athletes supported sharing their data with outside researchers in two scenarios. 15/16, 94%, said they would share their training load data with outside researchers when researchers asked each athlete directly for it. 13/15, 87%, said they would share their training data with outside researchers when researchers asked the team to provide athletes' data. A similar scenario with injury data, produced 15/15, 100%, and 13/15, 87%, respectively.

Table 4. Requirements for Downstream Data Sharing: Acceptable, Anonymization, Consent

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Training Load Data (Individual sharing with researchers)	Acceptable	●	●	○	●	●	●	●	●	●	●	●	●	●	●	●	●
	Consent	●	●	○	●	●	●	●	●	●	●	●	●	●	●	●	●
	Anonymization	○	○	○	●	○	●	○	●	○	●	●	○	○	●	○	●
Training Load Data (Team sharing with researchers)	Acceptable	●	●	●	●	●	●	●	●	●	-	●	●	●	●	●	●
	Consent	○	●	○	●	●	●	●	○	○	-	○	○	●	○	○	○
	Anonymization	○	○	○	●	○	○	○	●	○	-	●	○	●	●	○	●
Sleep Data (Individual sharing with other university teams)	Acceptable	○	○	○	●	○	○	○	●	○	-	●	○	●	●	○	●
	Consent	●	○	●	●	●	○	●	●	○	○	○	○	○	●	●	○
	Anonymization	○	●	○	●	○	●	●	○	●	●	●	●	●	○	○	●
Injury Data (Individual sharing with researchers)	Acceptable	-	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Consent	-	●	●	●	●	●	○	○	○	●	●	○	●	○	●	○
	Anonymization	-	○	●	●	○	●	●	●	●	○	●	○	●	○	○	●
Injury Data (Team sharing with researchers)	Acceptable	-	●	●	●	●	●	●	●	-	●	●	●	●	●	●	●
	Consent	-	●	○	●	●	●	○	●	-	●	○	●	●	○	●	●
	Anonymization	-	○	○	●	○	●	●	●	-	○	●	○	●	○	○	○
Data for Leaderboards (Individual sharing for Leader-board participation)	Acceptable	●	●	●	●	●	●	●	●	●	●	●	●	○	●	●	●
	Consent	○	○	○	●	●	○	○	○	○	○	●	○	●	●	●	○
	Anonymization	○	○	○	●	○	●	●	●	●	○	●	○	○	○	○	○

Student-athletes were asked if data sharing was acceptable in hypothetical contexts for different datatypes. A follow-up question then asked whether consent or anonymization should be applied when the datatype was going to be shared.

4.3 RQ3: Downstream Use

Rather than pin down a specific time frame, we considered any longer-term forward perspective to be a reflection of student-athletes' view of downstream use of their health and performance data.

4.3.1 Consent. A sample of interview questions had follow-up requests to specify what, if any, consent and anonymization the CI data type and CI sender should have in place. The wish to have consent or anonymization indicates some awareness for downstream use of that data. Results are shown in Table 4. In general, student-athletes are fine with sharing their data in these six examples. Greater than one-quarter of student athlete would like either consent (28/96) or anonymization (26/96) in place, and in a smaller percentage of instances, both consent and anonymization (15/96).

4.3.2 Personal Tracking. Sankey diagrams in Figure 3 show the information flows for athletes' sleep and non-sleep recovery, and for athletes' nutrition. Some participants show a "Personal" information flow. They told us that they have taken their own initiative to consult apps, personal journals, or outside human experts. The step to add information to help develop as an athlete may or may not be part of habit formation, but it does fall under our broader theme of Set up for Success. "If I know I have three hours of practice ahead, I know I have to eat. Thanks to my nutritionist, I know how to feel for that or like how I should feel for that," one student-athlete told us.

Most but not all of the student-athletes in the Women use case had an app or a device track their menstrual cycles (10/12). The role of information seeking in responses to questions about the purpose and benefit of tracking women athletes' menstrual cycles in college athletics was noticeable. For example one said, "I think it can help young women understand more like why they're feeling certain ways or like why they're less strong on on some days than others."

Menstruation was not listed as a subject of educational interest in SCAN, but a recent survey of top European women's soccer teams by the European Club Football organization reported[ref] that "76% expressed a willingness to learn more about the menstrual cycle." 72% of these soccer players felt that the menstrual cycle affects sports performance, a figure that is also consistent with our responses.

5 Discussion

College athletes are young people, in general and in our survey (average age of 20 years 8 months with standard deviation of 11.7 months). Ground truth or definitive conclusions are not realistic given our participants' formative stage of life. That does not mean that we cannot say things about privacy, and about the expectation that student-athletes have for privacy with the health and performance data that is used by their teams and by themselves. Contextual integrity is a framework for identifying contextual generalities and, with them, describing collective norms. The contrast between norms and individual opinions in the overarching college sports context reveal student-athletes' privacy expectations.

5.1 Highly Variable Interview Responses

The Training Load use case values in Table 1 are different from the other use case values. They are consistently lower than the values in the other use cases. The non-Training Load use cases display occasional, noticeable upward spikes in the I/We ratio values. The increases (more I, less We) indicate greater personal involvement with issues presented in a particular use case. Radar graph versions of Table 1 values are in Appendix A and they present a clearer picture of participants' response variation.

We use aggregated counts that describe support-levels (strong support, support, no support) for student-athletes' opinions as evidence to support conclusions from thematic analysis. We also rely on other surveys by the NCAA and by the European Club Football organization of similar athlete populations for additional evidence to support our conclusions. The supporting evidence bolsters the conclusions that have been developed through contextual integrity and thematic analysis.

The large variation of participants' I/We ratios and the broad set of participant responses to our questions are consistent with large student-athlete survey studies carried out by the NCAA, the organization that oversees college sports in the U.S. The NCAA carried out its 78-question GOALS (Growth, Opportunities, Aspirations, and Learning of Students in College) survey between December 2024 and June 2025. Likert-scale questions like "My head coach can be trusted" elicited 81% positive responses among men (69% for women) which is as close as GOALS comes to a consensus perspective.

The lack of consensus makes it difficult to bound contexts for student-athlete norms. That is not a bad thing, though it does make thematic analysis challenging. The opportunity is to seek out and find the fuzzy boundaries that lie in between contexts.

5.2 Tension between Athlete- and Team-centric

Our analysis resolves contradictory findings in Kolovson et al.[ref] which looked at power asymmetries between coaches and athletes and in Brewer et al.[ref] which looked at coaches' data relationships with student-athletes. Kolovson found that coach-athlete interaction can lead to outcomes not in the student-athlete's best interest. Brewer concluded that coaches "use data with the goal of balancing performance goals, athlete emotional well-being, and privacy."

The tension between Kolovson and Brewer is the tension we understand as the "gray areas" that we uncovered in relative thematic analysis. Our finding is that both of these types of athlete-coach interaction occur. Among the diverse perspectives of young student-athlete, the interaction between the individual athlete identity and the team social identity is identifiable, and can be a source of tension.

Each use case in the interview includes a hypothetical scenario. The hypothetical for the Training Load use case asked whether it was acceptable to share training load data with researchers working to create algorithms that predict future athletic performance. We asked athletes how they felt if their teams required them to follow algorithmic training recommendations based on their data, and how they felt if their teams shared their data with researchers without obtaining approval. Athletes, for the most part, supported their teams' decisions when it comes to collecting, using, and sharing training load data, as shown in Table 2.

Our interview script defines a setback as "a time when you were not able to reach your training goals," and we have a use case asking interview subjects questions about setbacks that they have experienced. The Setback use case has consistently higher I/We ratio numbers than the Training Load use case, as shown in Table 1. Two responses in Setbacks specifically call out the "gray area" between what's appropriate and what's inappropriate for information flow.

The hypothetical scenario in the Setbacks use case manufactures an athlete setback based on a coach's unfair, non-transparent key performance indicator (KPI), shown in Table 3a. Opinions were split in the interview responses to the hypothetical. Some student-athletes felt they needed a conversation to resolve the situation. Others said that they would do whatever they could to improve their KPI, even though they did know not what the path to improvement was. One said, "I don't think they're a good coach at all."

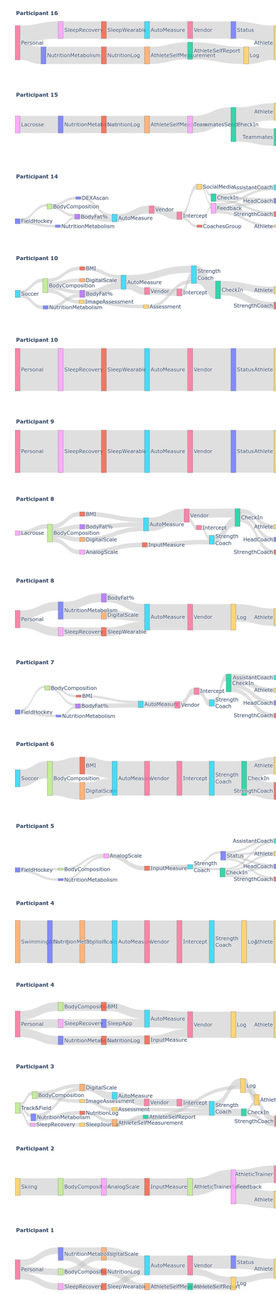


Fig. 3. Sankey diagrams showing combined team and personal information flows for tracking data of (1.) athletes' sleep recovery, (2.) athletes' non-sleep recovery, (3.) athletes' nutrition

5.3 Injuries, Habits, Success

The Injury use case in our study starts by asking student-athletes what "injury risk" and "injury prevention" mean to them. Risk is, by definition, the potential for harm, and prevention is how to avoid such harm. Our interviews did not talk about harm, and instead talked about habits. Injury prevention was their positive habits. Injury risk was whether or not they followed through with those habits.

We also asked student-athletes about their habits, in the Sleep Recovery use case and in the Nutrition use case. College athletes are aware of how important habits are for their own and their teams' success, but these are young people. Habit formation is an active learning process for them.

Only in hindsight did we realize that our injury questions about risk and prevention were invitations to student-athletes to express their confirmation bias. They would tell us what works for them to stay healthy and not injured. The presence of habits and confirmation bias in response to risk and harm provides a lens to evaluate information flows. If the habits are objectively positive then it is likely that information flows from data that informs those habits are also positive.

Sankey diagrams show the information flows for athletes' sleep and non-sleep recovery, and for athletes' nutrition (Figure 3). Some participants show a "personal" information flow. They told us that they have taken their own initiative to consult apps, personal journals, or outside human experts. The step to add information to help develop as an athlete may or may not be part of habit formation, but it does fall under our broader theme of Set up for Success. "If I know I have three hours of practice ahead, I know I have to eat. Thanks to my nutritionist, I know how to feel for that or like how I should feel for that," one student-athlete told us.

The most recent NCAA SNAP (Student-athlete Needs, Aspirations and Perspectives) study[ref] data from October 2025 asked 6800 NCAA Division I student-athletes[ref] across a range of subjects. One question on the SNAP survey asked student-athletes which topics of interest should have resources devoted to creating new educational materials. 56% of women and 41% of men said that nutrition is a subject of educational interest. 45% of women and 37% of men said that sleep is a subject of educational interest. Survey participation was voluntary.

The NCAA evidence further buttresses what we saw in our interviews, that student-athletes habit formation, personal initiative, and information seeking are important aspects of how they set themselves up for success.

Teams will point to injury prevention as a primary purpose of athletes' data[ref]. Student-athletes tell us that that purpose of injury-related data is to keep coaches, trainers and athletes all on the same page, and that coaches' benefit from better decision-making and from better caretaking of athletes. Ultimately, however athletes see injuries as the product of their habits. Habits are like other aspects of development that set up success, and they are more athlete-centric than team-centric.

5.3.1 Information Seeking. One revealing aspect of the responses was the role of information seeking in responses to questions about student-athletes' habits. One athlete pursued an independent study on how estrogen levels correlate with knee injuries, discovering that hormonal changes affect soft tissues' laxity over the course of women athletes' menstrual cycles.

The most recent NCAA SNAP (Student-athlete Needs, Aspirations and Perspectives) study[ref] data from October 2025 asked 6800 NCAA Division I student-athletes[ref] across a range of subjects. One question on the SNAP survey asked student-athletes which topics of interest should have resources devoted to creating new educational materials. 56% of women and 41% of men said that nutrition is a subject of educational interest. 45% of women and 37% of men said that sleep is a subject of educational interest. Survey participation in the NCAA SNAP study was voluntary.

5.4 Leaderboards

Leaderboards are the public, or semi-public display of athletes' data, in order to create intra-team competition, and with competition, to motivate the team's athletes. Leaderboards are an example of a self-contained information flow. Contextual integrity applies, and based on our student-athletes interviews we can fill in the framework parameters: subject, data type, sender, transmission principle, recipient. We will also cover the purpose and benefit of leaderboards, consistent with the CI framework[ref].

The sender for most leaderboard situations is the coach (head or assistant) who conceptualizes the data and its presentation. The subject is each athlete with data on the leaderboard and the data type is whatever data the coach/sender chooses. The presentation of the data in its leaderboard format is the transmission principle.

The purpose of a leaderboard is, at a surface level, motivation for athletes who, in many cases at the college-level, enjoy competition. Accountability among team members and athletes' education are by-products of the drills, or times, or whatever the reason for leaderboard competition happens to be. Recognition for athletes can sometimes be a factor. The coach/sender has lots of options with which to play up or play down the purpose aspects of a leaderboard.

Leaderboards can also arise as a result of their convenience. Coaches will unintentionally share athletes' data that they collect and use on a shared interface. For example, in one case, "When we report maxes and stuff we write them on a whiteboard and you can like, see who's leading, but it's never really advertised to us," according to one athlete. Student-athletes we spoke with admit to checking teammates' numbers in these situations and feeling competitive.

Individual sports' teams, like swimming and skiing, need leaderboards to function as selection criteria when a competition does not allow all of the athletes on a team to race. Many of these individual athletes also experience public versions of leaderboards, places where online aggregators track and post competition results. These online services have audiences that are CI receivers, and the sender is not the coach working in the college sports team context; it is the operator of the business.

5.5 Women Athletes' Data

The Womens use case focused on menstrual cycle tracking and women athletes. Interview responses had high I/We ratio values, indicating that athletes, for the most part, considered menstrual cycles more in its personal context than in its team context, even though the majority of female athletes felt that menstruation affected their athletic performance. Most but not all athletes used an app or a device to track their cycles (10/12).

Menstruation was not listed as a subject of educational interest in SCAN, but a recent survey of top European women's soccer teams by the European Club Football organization reported[ref] that "76% expressed a willingness to learn more about the menstrual cycle." 72% of these soccer players felt that the menstrual cycle affects sports performance, a figure that is also consistent with our responses.

When it comes to personal period tracking, 51% of NCAA SCAN responses and 69% of European soccer players reported tracking menstrual cycles. Our results are not out of line with these other reported percentages.

5.6 Roles, Responsibilities, Relationships

The word "friend" appears just one time in the 7 use cases across 16 transcribed interviews for this study. Student-athletes have relationships with coaches and teammates, but they are not bonding over their data. Some of that has to do with the power relationships where coaches oversee players, but the athletes we spoke to do not think in those terms.

Coaches' data responsibilities are not always executed to student-athletes' standards. The "endurance" CI data type from our questionnaire is shared among coaches, trainers, and teammates. The sharing is not to the extent of other team training CI data types (like fitness or strength), but it is greater than the more personal CI data types (like sleep or nutrition). A substantial percentage, 25%, of the access to endurance data was considered "not appropriate" in the survey. Teammates were most of those inappropriate instances, 5/7 or 71%.

Evidence based on the endurance data type sharing suggests that leaderboards sometimes happen when coaches' convenience supercedes athletes' privacy. Student-athletes do not appreciate the lack of discretion. Any biometric measurement can be applied to judge the personal character of an athlete, depending on how a group relates to the data[ref]. Endurance can problematic in that regard[ref]. Coaches are stewards of data, and stewards of team culture. Sometimes they get them both wrong.

Relationships between student-athletes and their collaborators have characteristic dynamics that also involve athletes' data, our thematic analysis showed. Team-centric dynamics, progression and consensus, bring a student-athlete together with coaches and teammates in ways that support planning and decision-making. Athlete-centric dynamics, adaption and maturity, reflect the student-athletes' experience of change and growth, helped by collaborators and by the evidence from their own health and performance data.

5.7 Privacy Expectations

Our research questions seek to define student-athletes' privacy expectations for three general data types: the ongoing tracking data, the data that is shared, and the downstream data given an uncertain future.

The gray areas where athlete-centric and team-centric information flows conflict occur in the present tense of ongoing data, like in an athletes' setback situation. This is our main finding. One could envision the conflict heightening as the scope of data sharing expands, but that is not something that we heard in our interviews. Student-athletes did tell us that they expect collaborators (coaches primarily) to be aware of and to respect personal boundaries for athletes' autonomy when it comes to acquiring and using athletes' data.

The student-athletes we spoke to take individual responsibility for their habits and other success behaviors. They willingly collaborate with the expert resources that their team provides (strength coach primarily) and ongoing tracking data is a central aspect of the collaboration. One could envision problems like the gray area conflict arising if student-athletes' autonomy in developing their own success behaviors gets compromised, but that is not something we heard in our interviews. Student-athletes expect to work with experts on their athletic development. Some also seek external experts and resources to augment what their team provides. In general, they are willing to have their data play an essential part of the process.

Leaderboards are an example of intra-team data sharing. Even though student-athletes are aware that not everyone on the team benefits from leaderboards, there are many reasons for having them. Opinions differed among the athletes we spoke to on whether the external motivation that a leaderboard competition fosters is a personal benefit. Based on what student-athletes told us about leaderboards, they expect data sharing to occur within their team and willingly go along with it.

We asked women athletes about data related to menstrual tracking. Even though most tracked their own period they felt the information was personal and not suitable to share with coaches and teammates. They did, however, acknowledge the potential benefit of sharing the data, even though they could understand how their team might benefit. Consistent with other women athlete groups, the student-athletes we spoke are information-seeking and curious about what future norms could look like. Student-athletes anticipate that norms could change with downstream data as technology, coaching, and training all change. They are not passive participants and with their information-seeking plan to play an active role in shaping those norms.

The student-athletes we spoke to recognize the different roles collaborators have in their athletic development. They defer to experts when it comes to decision-making and planning, and extend that deference to data sharing and to decisions about downstream data. Student-athletes expect coaches as part of their expertise to manage privacy in a responsible manner.

6 Conclusions

We applied the contextual integrity framework applied to the college sports context in an attempt to understand privacy norms and to report the privacy expectations of student-athletes. Privacy expectations are important to conceptualize, since CI posits that when when norms for shared information are respected and privacy expectations are met, then the contextual flows of information are appropriate. Our investigation finds that athlete- and team-centric information flows can be a source of tension, and that this tension affects privacy norms and expectations. Awareness of personal boundaries for student-athlete data by coaches and other collaborators is a coach's responsibility, according to student-athletes. Coaches need to navigate player boundaries and manage the tensions that arise in student-athletes' increasingly technology- and data-rich context.

The appropriate flow of student-athletes' health and performance data is one that balances athlete- and team-centric information flows and manages privacy in a responsible manner. Privacy-enhancing technology, like differential privacy, can simultaneously support the facets (norms, benefits, purposes) needed for appropriate information flow in a college sports context without engendering conflict, but proof-of-concept for that conjecture has not been demonstrated. (Work by TK comes the closest[ref] to proof-of-concept.) Still we believe that shareable yet private data give student-athletes the best opportunity to develop as athletes and to preserve the autonomy that helps them to develop as people.

7 Figures

The "`figure`" environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Your figures should contain a caption which describes the figure to the reader.

Figure captions are placed *below* the figure.

Every figure should also have a figure description unless it is purely decorative. These descriptions convey what's in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

A figure description must be unformatted plain text less than 2000 characters long (including spaces). **Figure descriptions should not repeat the figure caption – their purpose is to capture important information that is not already provided in the caption or the main text of the paper.** For figures that convey important and complex new information, a short text description may not be adequate. More complex alternative descriptions can be placed in an appendix and referenced in a short figure description. For example, provide a data table capturing the information in a bar chart, or a structured list representing a graph. For additional information regarding how best to write figure descriptions and why doing this is so important, please see https://www.acm.org/publications/taps/describing-figures/.

Fig. 4. 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (https://goo.gl/VLCRBB).

7.1 The "Teaser Figure"

A "teaser figure" is an image, or set of images in one figure, that are placed after all author and affiliation information, and before the body of the article, spanning the page. If you wish to have such a figure in your article, place the command immediately before the `\maketitle` command:

```
\begin{teaserfigure}
  \includegraphics[width=\textwidth]{sampleteaser}
  \caption{figure caption}
  \Description{figure description}
\end{teaserfigure}
```

8 Citations and Bibliographies

The use of BibTeX for the preparation and formatting of one's references is strongly recommended. Authors' names should be complete — use full first names ("Donald E. Knuth") not initials ("D. E. Knuth") — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two commands, placed just before the `\end{document}` command:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{bibfile}
```

where "`bibfile`" is the name, without the "`.bib`" suffix, of the BibTeX file.

Citations and references are numbered by default. A small number of ACM publications have citations and references formatted in the "author year" style; for these exceptions, please include this command in the **preamble** (before the command "`\begin{document}`") of your LaTeX source:

```
\citestyle{acmauthoryear}
```

Some examples. A paginated journal article [2], an enumerated journal article [11], a reference to an entire issue [10], a monograph (whole book) [24], a monograph/whole book in a series (see 2a in spec. document) [18], a divisible-book such as an anthology or compilation [13] followed by the same example, however we only output the series if the volume number is given [14] (so Editor00a's series should NOT be present since it has no vol. no.), a chapter in a divisible book [36], a chapter in a divisible book in a series [12], a multi-volume work as book [23], a couple of articles in a proceedings (of a conference, symposium, workshop for example) (paginated proceedings article) [3, 16], a proceedings article with all possible elements [35], an example of an enumerated proceedings article [15], an informally published work [17], a couple of preprints [6, 8], a doctoral dissertation [9], a master's thesis: [4], an online document / world wide web resource [1, 28, 37], a video game (Case 1) [27] and (Case 2) [26] and [25] and (Case 3) a patent [34], work accepted for publication [31], 'YYYYb'-test for prolific author [32] and [33]. Other cites might contain 'duplicate' DOI and URLs (some SIAM articles) [22]. Boris / Barbara Beeton: multi-volume works as books [20] and [19]. A presentation [30]. An article under review [7]. A couple of citations with DOIs: [21, 22]. Online citations: [37–39]. Artifacts: [29] and [5].

9 Acknowledgments

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document.

This section has a special environment:

```
\begin{acks}
...
\end{acks}
```

so that the information contained therein can be more easily collected during the article metadata extraction phase, and to ensure consistency in the spelling of the section heading.

Authors should not prepare this section as a numbered or unnumbered `\section`; please use the "`acks`" environment.

A Appendix: Interview Materials

Primer: Screening Details and Definitions

Confirm details from screening

Printout has screening survey results (age, college eligibility year, sport) and important definitions and 4 initial Foundational Questions
Question: Are all of the answers in your screening questionnaire correct?

Definitions in this interview

Performance Tracking Technology (PTT) - These are sensor and computer technologies, often either wearable or computer-vision based, that collect athletes' health and performance data at regular intervals for analysis of change over time.

Appropriateness - This is a subjective judgement of whether something is suitable given its circumstance. The opposite of appropriate is inappropriate. In the context of personal data and privacy, which is what we are studying, appropriate refers to information flows that are considered acceptable and permissible within specific social and technical situations.

Conflicting Interests - There can be times when the best interests of teams, coaches and athletes are at odds, and those circumstances extend to PTT. Your answers to our questions should reflect your self-interest, but please mention it if your self-interest conflicts with the interests of your team or your coach.

Athlete Development and Biopsychosocial Model - We apply the premise that athletic development involves facets that are physiological, mental and involve interpersonal relationships.

Competitive Advantage - One assumption that we make is that competitive advantage is the primary motivation for team-oriented decision-making among athletes, coaches and any other staff on the team. Athletes also seek competitive advantage personally in their intra-team competition with other athletes.

Training Load - Cumulative amount of strain on the body from exercise as determined by PTT.

Head Coach, Training Staff, Assistant Coaches - Plural “Coaches” refers to Head Coach and all Assistant Coaches collectively. Training staff would be separate from coaches if these people have singular responsibility for athletes’ fitness and training without the skill development that coaches normally undertake.

KPI (Key Performance Indicator) - KPIs distill PTT and other athlete data to succinctly describe athletes’ performance and health status and athletes’ development progress.

Confirm: Do you have any questions about any of the definitions? Do you have questions about anything so far before we start on the interview?

Reminder: You have agreed to record our interview. You can pause or stop the interview at any time

Check all boxes that apply. **Health & performance data types in our team**	Check all boxes where True. **This data has helped me to become a __________ .**	Check all boxes that apply. **Who has access to use your data?**	Check all boxes where True. **It is appropriate for ______ to access this data.**
Fitness Testing Data ☐ GPS for sprints ☐ Counter movement jump ☐ Velocity-based strength ☐ Heart rate	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Training Load Data ☐ Position tracking ☐ Strength ☐ Perceived exertion ☐ Heart rate ☐ Psycho-social stress	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Skill Development Data ☐ Time on task ☐ Video ☐ Accelerometer - IMU ☐ Biomechanics ☐ Ball/Object tracking ☐ Reaction time	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Movement/Symmetry Data ☐ Functional movement ☐ Computer vision ☐ Motion capture ☐ Lidar ☐ Observation	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Body Composition Data ☐ Analog scale ☐ Digital scale ☐ Body fat percentage ☐ Body mass index (BMI) ☐ DXA scan ☐ Photo/Video assessment ☐ 3D muscle volume	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Strength Training Data ☐ Free weights lifts log ☐ Bodyweight positions ☐ Isokinetic dynamometer ☐ Velocity-based training ☐ Auto rep & set counter	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Jump Training Data ☐ Plyometrics, Jump squats ☐ Vertec measurement ☐ Vertimax ☐ Motion capture ☐ Biomechanics analysis ☐ Force plate	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates

Check all boxes that apply. **Tracked Data Type & Associated PTT**	Check all boxes where True. **This data has helped me to become a __________ .**	Check all boxes that apply. **Who has access to use your data?**	Check all boxes where True. **It is appropriate for ______ to access this data.**
Head Impacts Data ☐ Helmet sensors ☐ Mouthguard sensors ☐ Video analysis ☐ Concussion symptoms ☐ Observation	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Speed Training Data ☐ Position tracking ☐ Motion Capture ☐ Biomechanics ☐ Video ☐ Treadmill	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Endurance Training Data ☐ GPS/tracking distance ☐ Training intensity ☐ VO2max ☐ Lactate threshold ☐ Core body temperature	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Sleep Recovery Data ☐ Wearable sensor + algorithms ☐ Mattress sensor + algorithms ☐ Sleep journal ☐ Polysomnography	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Non-sleep Recovery Data ☐ Refueling nutrition ☐ Hydration ☐ Normatec/Message ☐ Active recovery ☐ Total Quality Recovery	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Nutrition/Metabolism Data ☐ Scale ☐ Body-fat measurement ☐ Nutrition log or tracker ☐ Continuous glucose monitor ☐ Breathing sensor	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates
Fluid Balance/Hydration Data ☐ Sweat biomarkers sensor ☐ Urine color analysis ☐ Sweat loss by weight ☐ Drink at thirst & monitor	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates

Foundational Questions

Most of the questions in this interview are specific examples and usage cases for performance tracking technology (PTT). Before we start on the specifics we have some general inquiries.

Balancing athletic, academic, and other commitments

We have a biopsychosocial perspective on athlete development.It means that your physical health, your mental wellbeing and your interpersonal relationships all contribute to your athletic success.

1. Compared to non-athlete students, do you have a harder or easier time meeting your academic and social commitments? Why?
2. Compared to teammates, do you have a harder or easier time meeting your athletic, academic and social commitments? Why?

Athlete-coach personal interactions

1. What are your one-on-one interactions with your coaches like?
2. Do your coaches mention PTT in one-on-one interactions always/sometimes/never?
    a. If always or sometimes, what are those interactions like?

Appropriateness

1. Do you feel comfortable assessing whether different types of PTT are appropriate or inappropriate?
2. Do you have any questions about what we mean when we ask you to tell us if something is appropriate?

General PTT Perspective

1. In general, what do you think about college athletics teams collecting and sharing student athletes' health and athletic performance data?
    a. In your opinion, is too much or too little or just the right amount of PTT data collected in your team?
    b. In your opinion, is there too much or too little or just the right amount of PTT data that your coaches and trainers share back with you?

Usage Case 1: Training Load

Copy Answers from **PTT Questionnaire #5** (PTTQ#5)

Training Load Data ☐ Position tracking ☐ Strength ☐ Perceived exertion ☐ Heart rate ☐ Psycho-social stress	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates

Personal Sensemaking & Experience

1. How would you define training load?

Current Team Practices

1. If PTTQ#5 checked:Describe how training load data is used by ______ [checks in column 3].
    a. If "Teammates" checked PTTQ#5 third column: Are there situations where Teammates use training load data without coaches or training staff?
2. If PTTQ#5 not checked: Take 1 minute and think, are there any questions about training load that we can answer?
    a. Under what circumstances would you want to have training load data collected?

Purpose & Benefits

1. In your opinion, what is the purpose of tracking training load in your team?
2. In your opinion, who benefits from tracking training load data? Why and How?

Information Flow (Sharing and Use)

1. Is there anyone not listed who has access to your training load data?
2. Among the people who access your training load data, are there differences in how they use it?
    a. If yes, please describe the differences.
    b. Do the differences matter in determining whether their access is appropriate or not? Why or why not?

Hypothetical Scenario: A group of not-for-profit sports science researchers develop algorithms that can reliably predict future athletics performance based on current training load data.

1. The algorithms can provide individualized recommendations on training load targets to improve their athletic ability. The researchers make it available to your team (including you) at no cost.
    a. As an individual athlete, would you follow the individualized recommendation? Why?
    b. How would you feel if your team required you to follow the recommendation? Why?
2. Let's say these sports science researchers need more data to improve the algorithms:
    a. As an individual athlete, **would you share** your training load data with the researchers? Follow-up: What (1) anonymization and (2) consent practices would you prefer?
    b. Is it acceptable for **your team** to share your training load data with the researchers? Follow-up: What (1) anonymization and (2) consent practices would you prefer?

Usage Case 2: Sleep Recovery

PTTQ#11

<u>Sleep Recovery Data</u> ☐ Wearable sensor + algorithms ☐ Mattress sensor + algorithms ☐ Sleep journal ☐ Polysomnography	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates

Personal Sensemaking & Experience

1. Based on your personal experience, does sleep affect your athletic performance?

Current Team Practices

1. Follow up to PTTQ#11 column 1
   a. [If checks] Do you think that team-wide sleep tracking helps your team? Why?
   b. [If no checks] Do you think that team-wide sleep tracking would help your team? Why?
2. Follow up to PTTQ#11 column 2
   a. [If checks] Please explain why and how. (sleep tracking makes you Better _____) ?
   b. [If no checks]: Please explain why.

Purpose & Benefits

1. In your opinion, what is the purpose of tracking sleep in your team/college sports teams?
2. In your opinion, who benefits from tracking sleep data in college sports? Why and how?

Information Flows (Sharing and Use)

1. Follow up to PTTQ#11 column 4
   a. [Both checks/no checks]: Could you please explain why it is (would be) appropriate/inappropriate for ___ to access your sleep tracking data?
2. Are there circumstances that limit when or if sharing sleep tracking data is appropriate?

Hypothetical Scenario: Let's say that different teams at UVM track athletes' sleep data. The coaches intend to share the data among themselves to analyze how sleep data impact athletic performance at the UVM athletics level.

1. How do you feel about the data sharing and use practices in the scenario?
   a. Is it appropriate? Why and how?
   b. If this is happening, what (1) anonymization and (2) consent practices would you prefer?

Usage Case 3: Injury Prevention, Risk and Occurrence

Show participants' answer to Screening Questionnaire #6 (SQ#6): Have you experienced a significant injury, one that forced you to miss two or more consecutive games during your high school or college season?

Personal Sensemaking & Experience

1. Based on your experience, how do you think performance tracking relates to athlete injury occurrence?
    a. What does "injury prevention" mean to you?
    b. In your opinion, what role does PTT play in injury prevention?
    c. What does "injury risk" mean to you?
    d. In your opinion, what role does PTT play in managing injury risk?
2. Have you ever experienced an injury that was difficult to diagnose or figure out?
    a. If yes, what was the experience like? And what role did PTT play in understanding the injury?

Current Team Practices

1. Is any of your team's training and PTT directed specifically at injury prevention?
    a. If yes, what?
    b. If no, what should your team do for injury prevention?
2. Does your team do injury post-mortems or after-injury analysis?
    a. If yes, what is it like?
    b. If yes, what role does PTT play in after-injury analysis?
3. How does your current team keep track of athlete injury occurrences?
4. Does your team use PTT to keep track of athlete injury risk? If yes, how?
5. On your team, who has access to athletes' injury-related data?

Purpose and Benefit

1. In your opinion, what is the purpose of tracking injury-related data in college sports teams?
2. In your opinion, who benefits from tracking injury-related data? Why and How?

Information Flows (Sharing and Use)

1. In your opinion, is it appropriate for [medical personnel/coaches/training staff/teammates/yourself] to have access to your injury-related data?
    a. Medical personnel: appropriate [ ] inappropriate [ ]
    b. Coaches: appropriate [ ] inappropriate [ ]
    c. Training staff: appropriate [ ] inappropriate [ ]
    d. Teammates: appropriate [ ] inappropriate [ ]
    e. Yourself: appropriate [ ] inappropriate [ ]
    f. Please explain why it's appropriate or inappropriate for ____ to access this data?

Hypothetical scenario: Let's say that a group of not-for-profit sports science researchers want to develop algorithms that analyze their PTT data. athletes of elevated injury risks

2. Let's say these sports science researchers need more data to improve the algorithms:
    a. As an individual athlete, **would you** share your injury-related data with the researchers? Follow-up: What (1) anonymization and (2) consent practices would you prefer?
    b. Is it acceptable for **your team** to share your injury-related data with the researchers? Follow-up: What (1) anonymization and (2) consent practices would you prefer?

Usage Case 4: Athlete Development Setbacks

Personal Sensemaking & Experience

1. Can you recall a time when you were not able to reach your training goals (defined as setbacks)?
    a. Follow-up: How did you know that you were experiencing a setback? With any type of PTT?
    b. How did you overcome the setback? With any type of PTT?
2. In general, what are the lessons you learned in the process of overcoming athletic setbacks?
    a. What have you learned from or about PTT in the experience? Does any of the PTT learning extend to coaches or teammates?

Current Team Practices

1. In your team, how do coaches and training staff manage athletes who experience setbacks?
    a. Follow-up: Does PTT play a role in the current team practice? How?

Purpose and Benefits

1. In your opinion, does PTT help avoid or overcome setbacks in athlete development? How?
2. Does PTT become more or less important to you when you're experiencing a setback? Why and how?
3. In a setback situation, who benefits from PTT whether or not you succeed? Why and how?

Information Flows (Sharing and Use)

1. How do you feel about college teams using PTT data to manage athlete development setbacks?
    a. Follow-up: Are there specific purposes or contexts that make it appropriate/inappropriate?

Hypothetical Scenario: Let's say your coach relies on a key performance indicator (KPI) score to gauge athletes' fitness. The KPI is calculated by combining multiple PTT data sources and the calculation is not transparent to athletes.

1. Is it appropriate for a coach to use such a KPI from athletes' PTT data? Why or why not?
2. Do you think a single KPI score is an accurate assessment of athletes' fitness? Why or why not?
3. Let's say that your coach says that your role on the team is going to be reduced, and cites a decline in your KPI as the primary reason. What would you do to improve your situation?
4. How do you figure out what's best for you AND what's best for the team when PTT presents conflicting information?

Usage Case 5: Leaderboard

Personal Baseline:

1. Compared to your teammates, how competitive would you say you are in your sport? (1) more competitive than teammates (2) similarly competitive as my teammates (3) less competitive than teammates

Current Team Practices

1. Does your team use leaderboards?
    a. If yes: Tell us who uses it, how they are used, and what information is on it.
    b. If no: In your opinion, why doesn't your team use it?

Purpose and Benefits

1. In your opinion, what's the purpose of leaderboards in college athletics?
    a. Follow-up: How well does it serve its purpose in your team?
2. In your opinion, who benefits from leaderboards in college athletics? How?

Information Flows (Sharing and Use)

1. Is it appropriate for college athletic teams to use leaderboards?
    a. Follow-up: are there specific purposes or contexts that make it inappropirate?
2. In your opinion, do leaderboards show information that is not appropriate?
    a. What information? Why?

Hypothetical Scenario: A coach and a trainer have a leaderboard that shows players' speed from location tracking data. The trainer is careful to point out that the leaderboard describes speed in a singular context, and the goal is to improve players' performance in those situations.The coach wants to spark competition throughout the team, and points out that the athletes at the bottom are "holding the average down."

1. What, in your opinion, is different about the two motivational approaches?
2. **How appropriate is each motivational approach?**
3. Given the opportunity, what would you like to see in terms of anonymizing the leaderboard data?
4. What, if any, consent should the trainer and coach get from players before creating their leaderboard?
5. The leaderboard only shows athletes who have given permission so you then know who among your teammates does not want their effort on display. In your opinion, are there repercussions for choosing not to participate?
6. A mental health advisor from the athletic department sees the effort leaderboard and identifies an athlete who might need counseling. What's appropriate for the advisor when it comes to possibly intervening with an at-risk athlete?

Usage Case 6: Nutrition & Body Composition

PTTQ#13

Nutrition/Metabolism Data ☐ Scale ☐ Body-fat measurement ☐ Nutrition log or tracker ☐ Continuous glucose monitor ☐ Breathing sensor	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates

PTTQ#6

Body Composition Data ☐ Analog scale ☐ Digital scale ☐ Body fat percentage ☐ Body mass index (BMI) ☐ DXA scan ☐ Photo/Video assessment ☐ 3D muscle volume	☐ Better Athlete ☐ Better Student ☐ Better Teammate	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates	☐ You, the athlete ☐ Head Coach ☐ Training Staff ☐ Assistant Coaches ☐ Teammates

Personal Sensemaking and Experience

1. Based on your experience, does nutrition & body composition affect your athletic performance?

Current Team Practice

1. Does your team provide free snacks and/or meals to players?
    a. If yes, are you aware of any nutritional guidance or tracking related to team-provided snacks or meals?
        i. If yes, what is the extent?
2. Does your team have a nutritionist on staff?
    a. If yes, to the extent that you are aware, what player information is available to them?
3. [If PTTQ#13 & PTTQ#6 checks] Does your team set goals for athletes' nutrition & body composition data? If yes, how
    a. Does your team link athletes' nutrition & body composition data to their athletic training and performance? If yes, how?

Purpose and Benefits

1. In your opinion, what is the purpose of tracking (1) nutrition data and (2) body composition data in college athletics?
2. What information is appropriate for a team nutritionist to have?
3. In your opinion, what are the benefits from (1) nutrition data and (2) body composition data in college athletics? How?
4. [If PTTQ#13, PTTQ#6 checks] Could you explain how the tracking benefits you?
5. In your opinion, are there downsides for tracking (1) nutrition data and (2) body composition data?

Information Flows (Sharing and Use)

3. Follow up to PTTQ#13 column 3
    a. [If checks] What do you think that _______ access your sleep tracking data?
    b. [If no checks]: Would it be appropriate for ___to access your sleep tracking data?
    c. [Both checks/no checks]: Could you please explain why it is (would be) appropriate/inappropriate for ___ to access your sleep tracking data?

Hypothetical Scenario: You have a 1-on-1 meeting with the team nutritionist who wants to tell you that your body composition has changed and your body fat percentage has increased. The nutritionist is well-respected and has contributed to championship teams. You know that your body fluctuates and you have not sensed anything out of the ordinary.

- The nutritionist feels like it's important information, that it will help you and the team to reach its goals, but also wants to protect your emotional well-being. Is the nutritionist right to be concerned? Why or why not?
- PTT data shows that your fitness is excellent despite the body change. Do you agree or disagree that the body change is important information given your fitness? Why?
- You said that it is appropriate for some (or no) people to access this data. Is it appropriate for the team's nutrition or medical staff to access this data?

Usage Case 7: Women Athlete Questions

We have questions that are unique to women athletes. You don't have to answer every question if you don't feel comfortable. Reminder, this study is anonymous and your answers will help us understand these important topics to eventually improve the wellbeing of women athletes.

Personal Sensemaking and Experience

1. Research has shown that menstrual cycles impact women's athletic performance. Can you share your personal experience about how your menstrual cycles affect your athletic performance?

Current Personal/Team Practice

1. Do you track your own cycles?
    a. If yes: How?
2. Does your team track women athletes' cycles?
    a. If yes: How? Who has access to the data?

Purpose and Benefits

1. In your opinion, what is the purpose of tracking women athletes' menstrual cycle in college athletics?
2. In your opinion, who benefits from tracking women athletes' menstrual cycle in college athletics? Why and how?

Information Flows (Sharing and Use)

1. How appropriate do you feel about college athletic teams tracking menstrual cycle data?
    a. Are there any specific tracking practices appropriate?
    b. Does the purpose of the tracking affect the appropriateness?
2. In the college athletics context, it's appropriate for ___ to have access to your cycle data
    b. Recipients: self, head coach, other team coaches, training staff, teammates
    c. Does the gender of the person affect the appropriateness?
3. Do you have other concerns or comments about college athletic teams tracking this data

Hypothetical scenario: Apart from tracking menstrual cycles, some PTT we discussed today can be used to indirectly infer women athletes' cycles from other performance or biometric data.

1. Do you have any first-hand experience with this?
2. How appropriate is it for the team to infer women athletes' menstrual cycles from other PTT data?
3. Do you have concerns or comments about the inferred cycles from PTT data?

B Appendix: Exploratory Radar Graphs

womens data

○ yes ◉ no

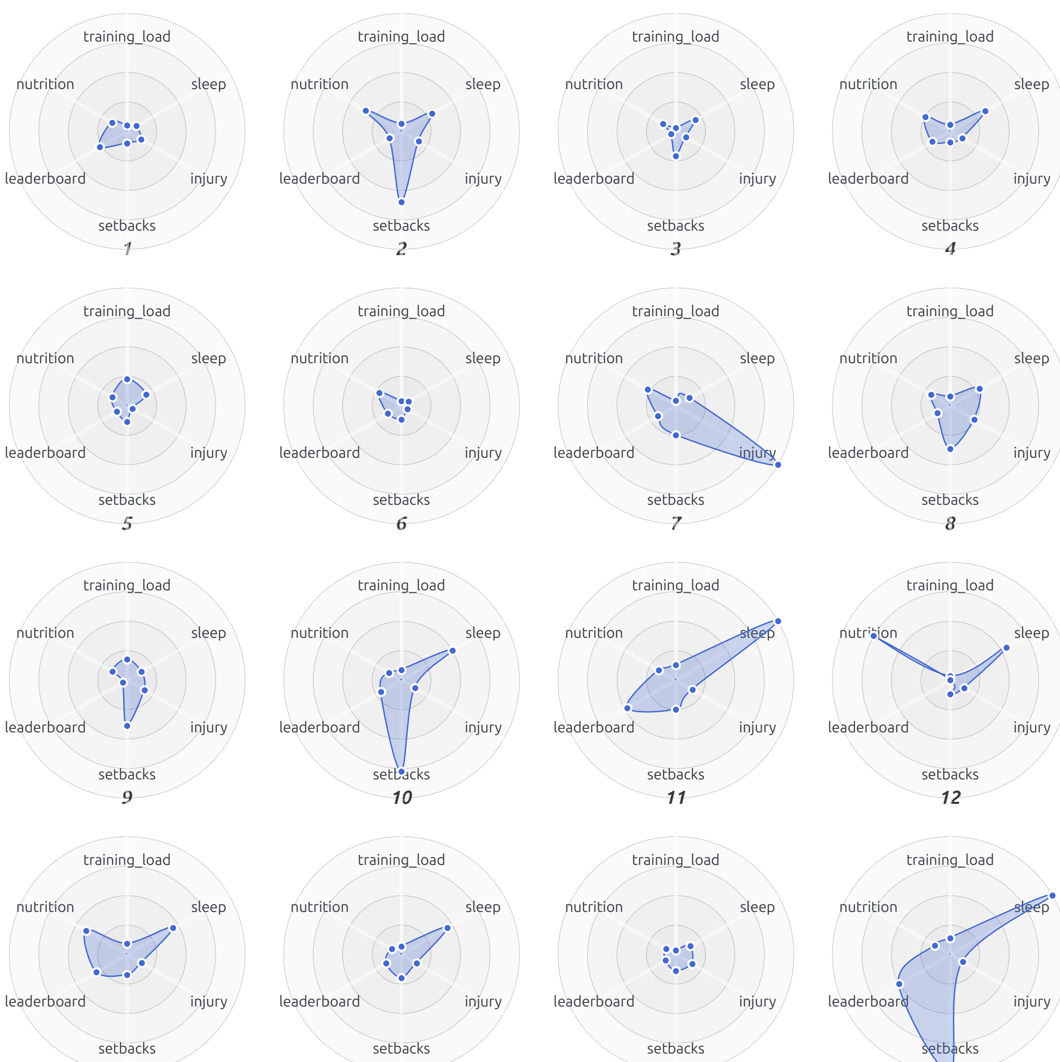
identifier

☑ id ☐ team ☐ gender ☐ age ☐ iot ☐ injuryYN

sort by parameter

default

▶ Code



womens data

○ yes ◉ no

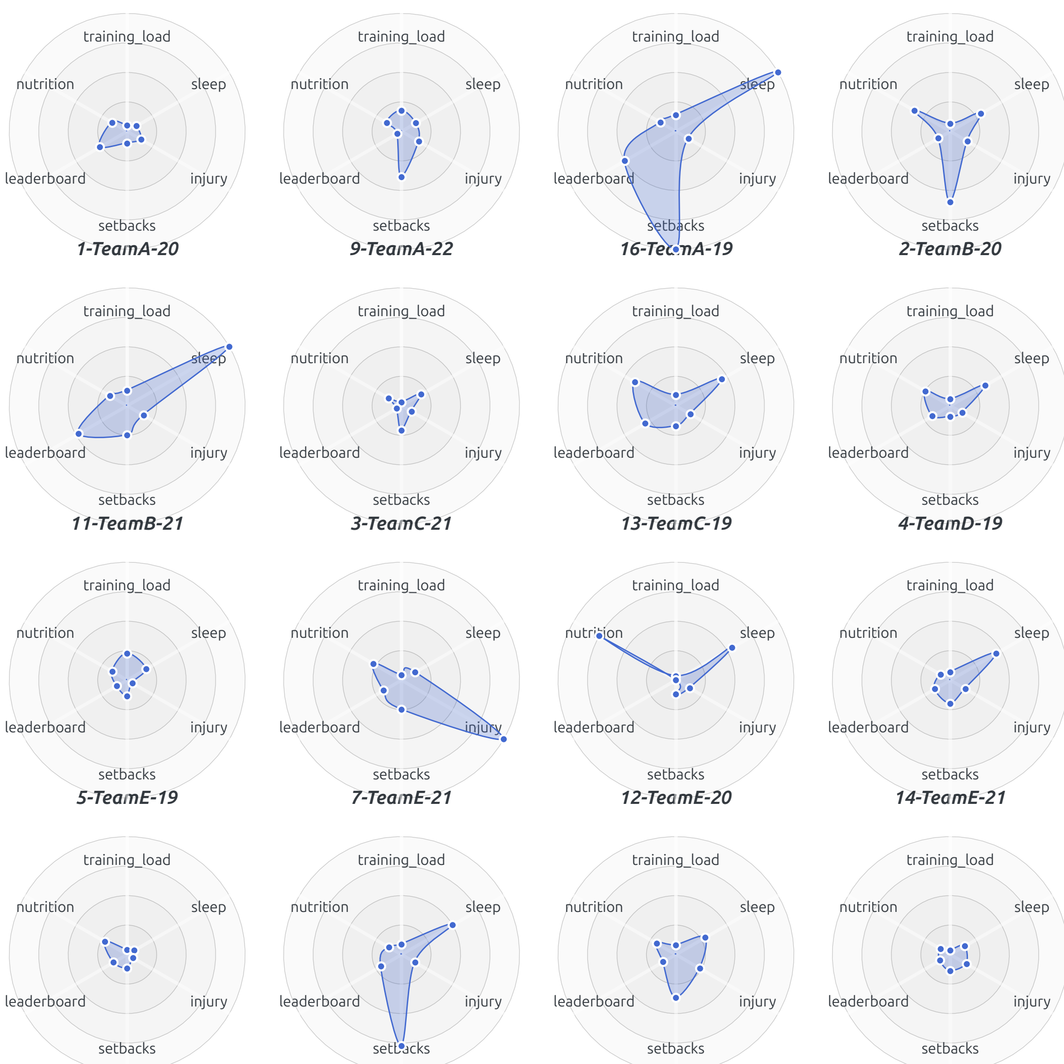
identifier

☑ id ☑ team ☐ gender ☑ age ☐ iot ☐ injuryYN

sort by parameter

which team

► Code



womens data

◉ yes ○ no

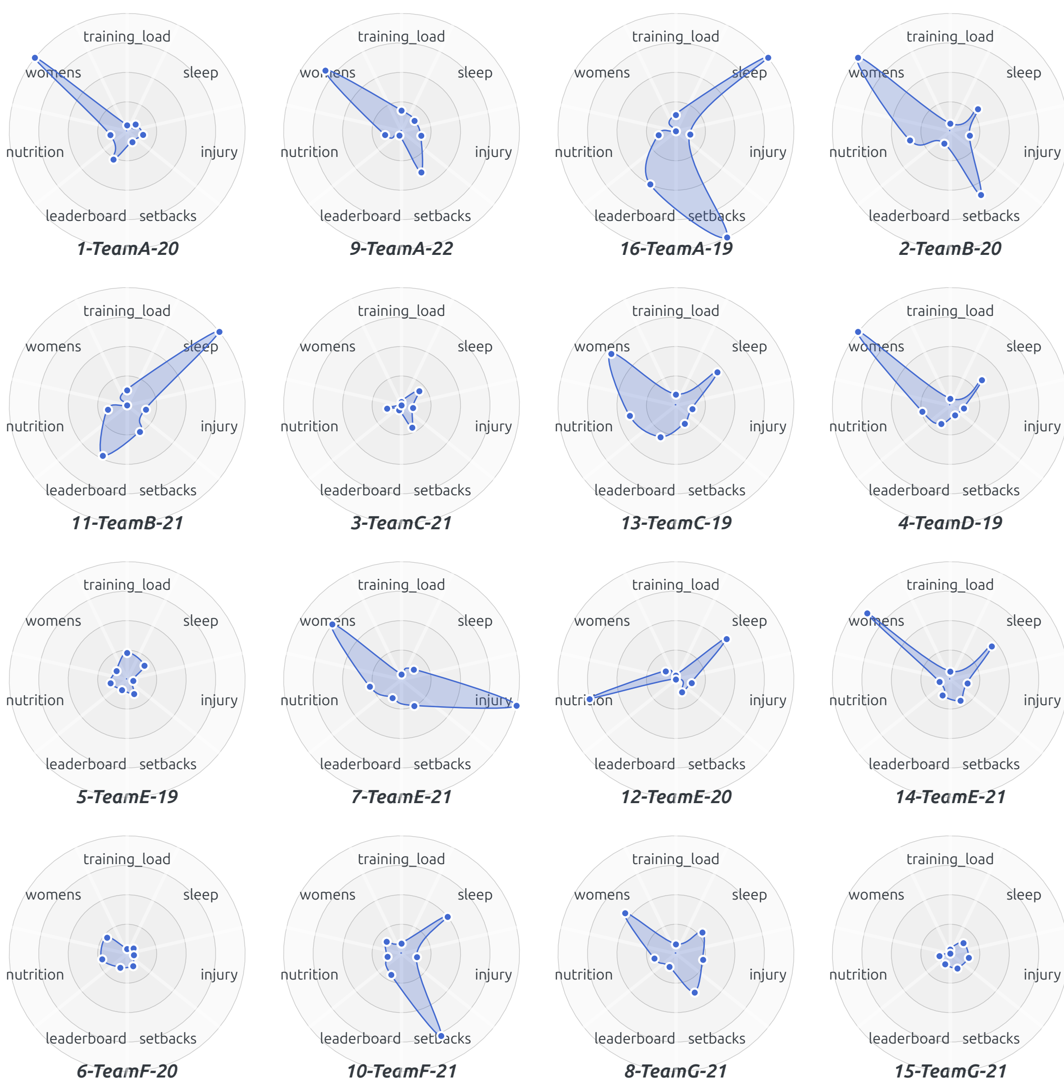
identifier

☑ id ☑ team ☐ gender ☑ age ☐ iot ☐ injuryYN

sort by parameter

which team

▸ Code



C Appendix: Information Flow Diagrams

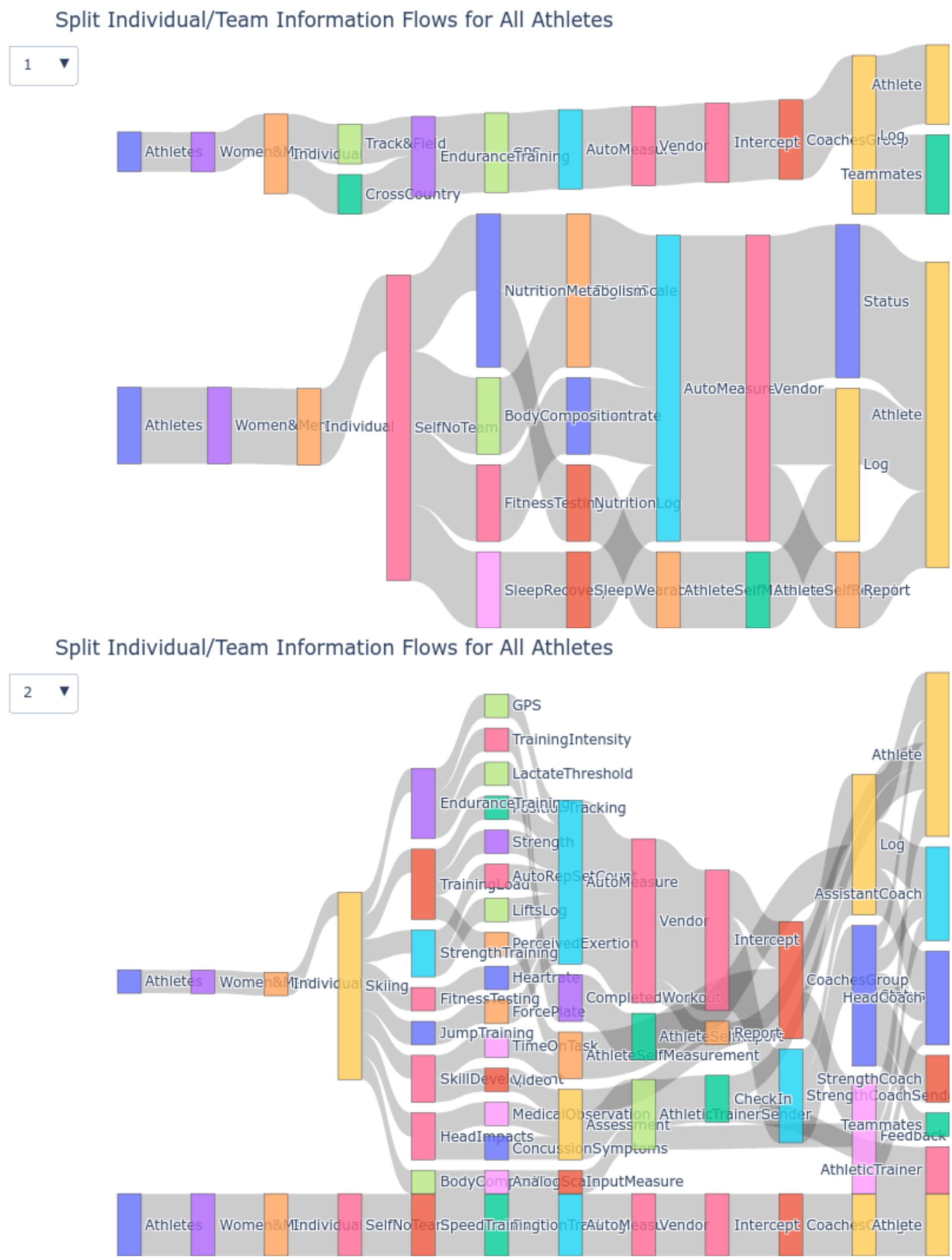


Fig. 5. Sankey Diagrams for Directed Graph Contextual Integrity Information Flows

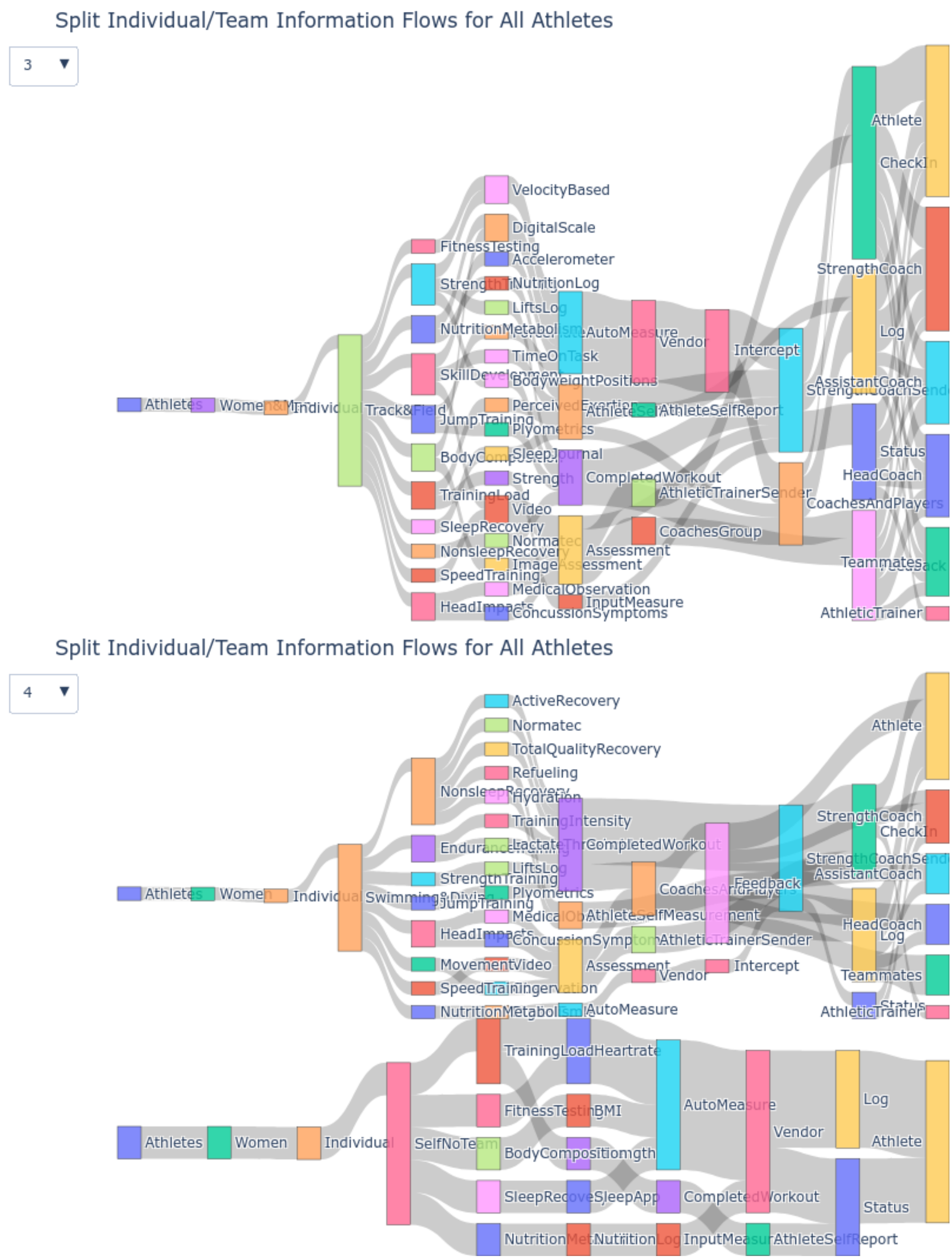


Fig. 6. Sankey Diagrams for Directed Graph Contextual Integrity Information Flows

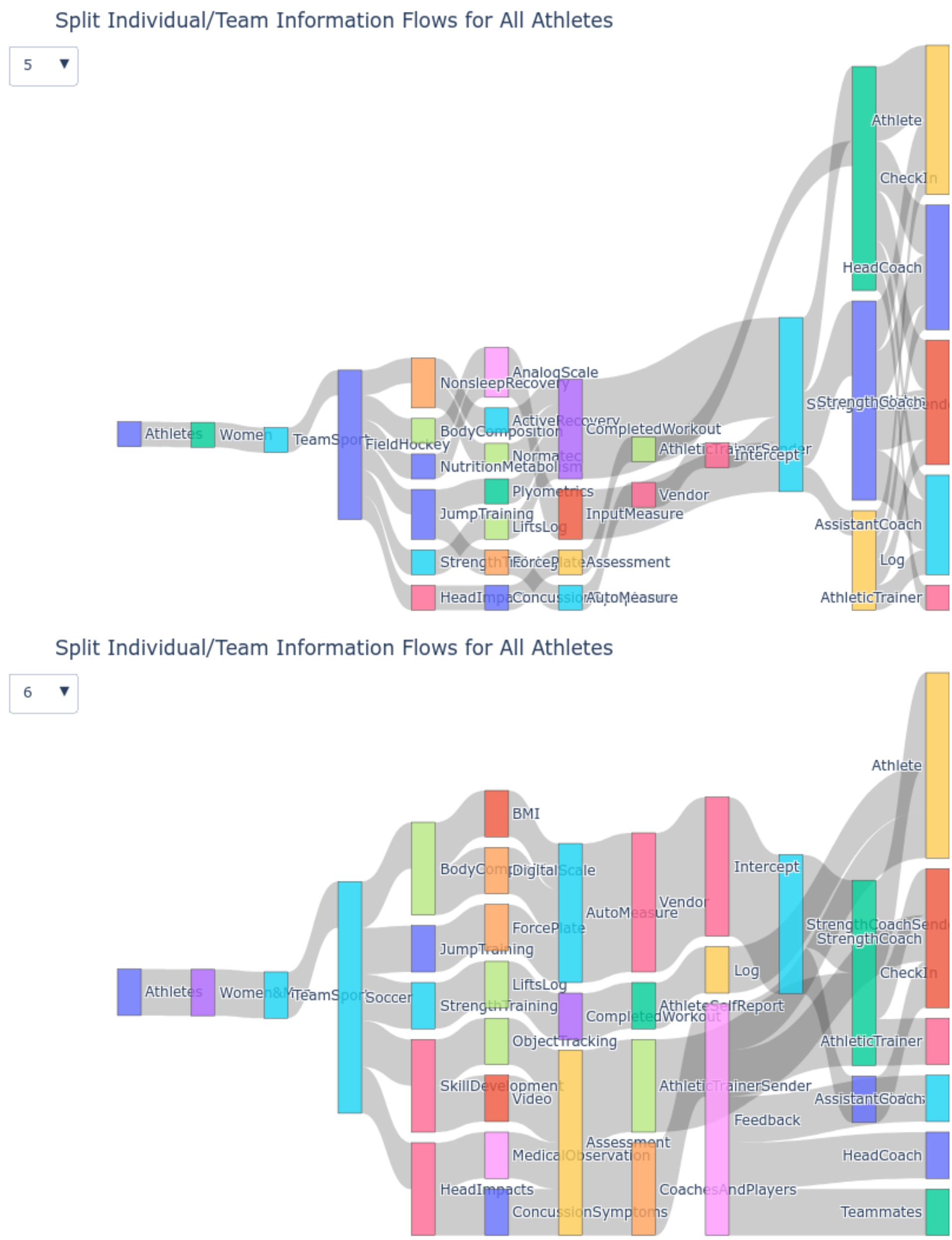


Fig. 7. Sankey Diagrams for Directed Graph Contextual Integrity Information Flows

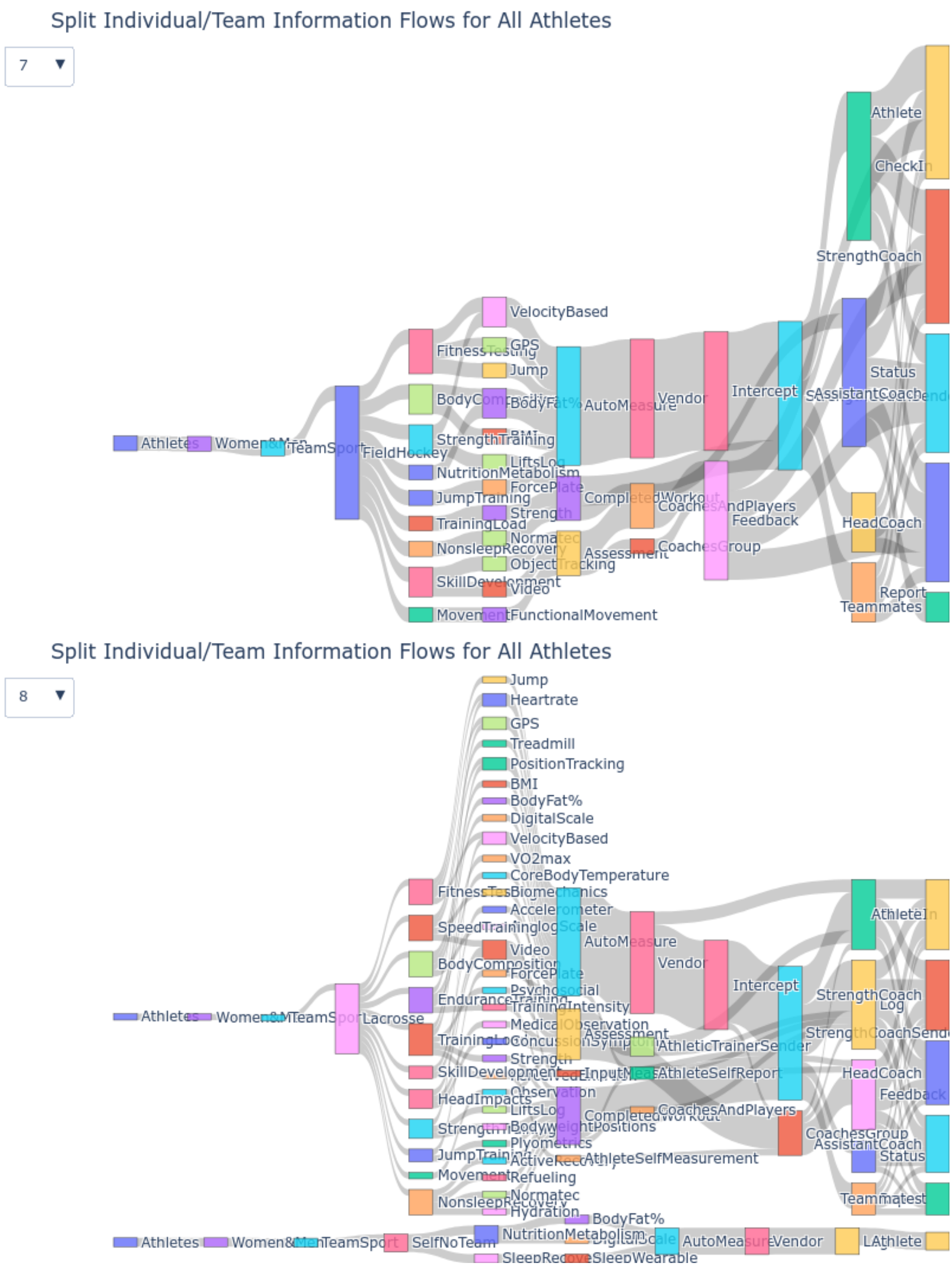


Fig. 8. Sankey Diagrams for Directed Graph Contextual Integrity Information Flows

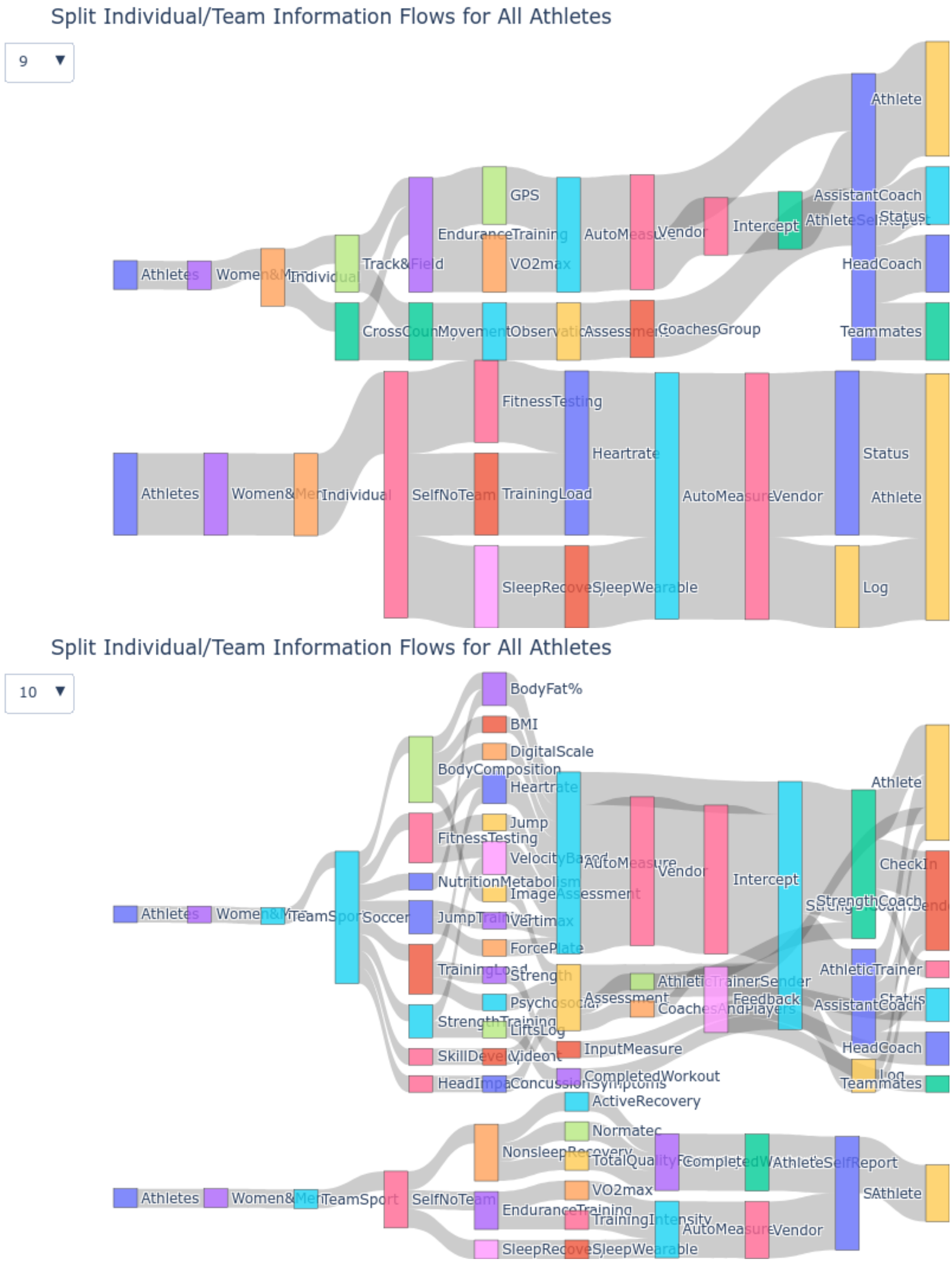


Fig. 9. Sankey Diagrams for Directed Graph Contextual Integrity Information Flows

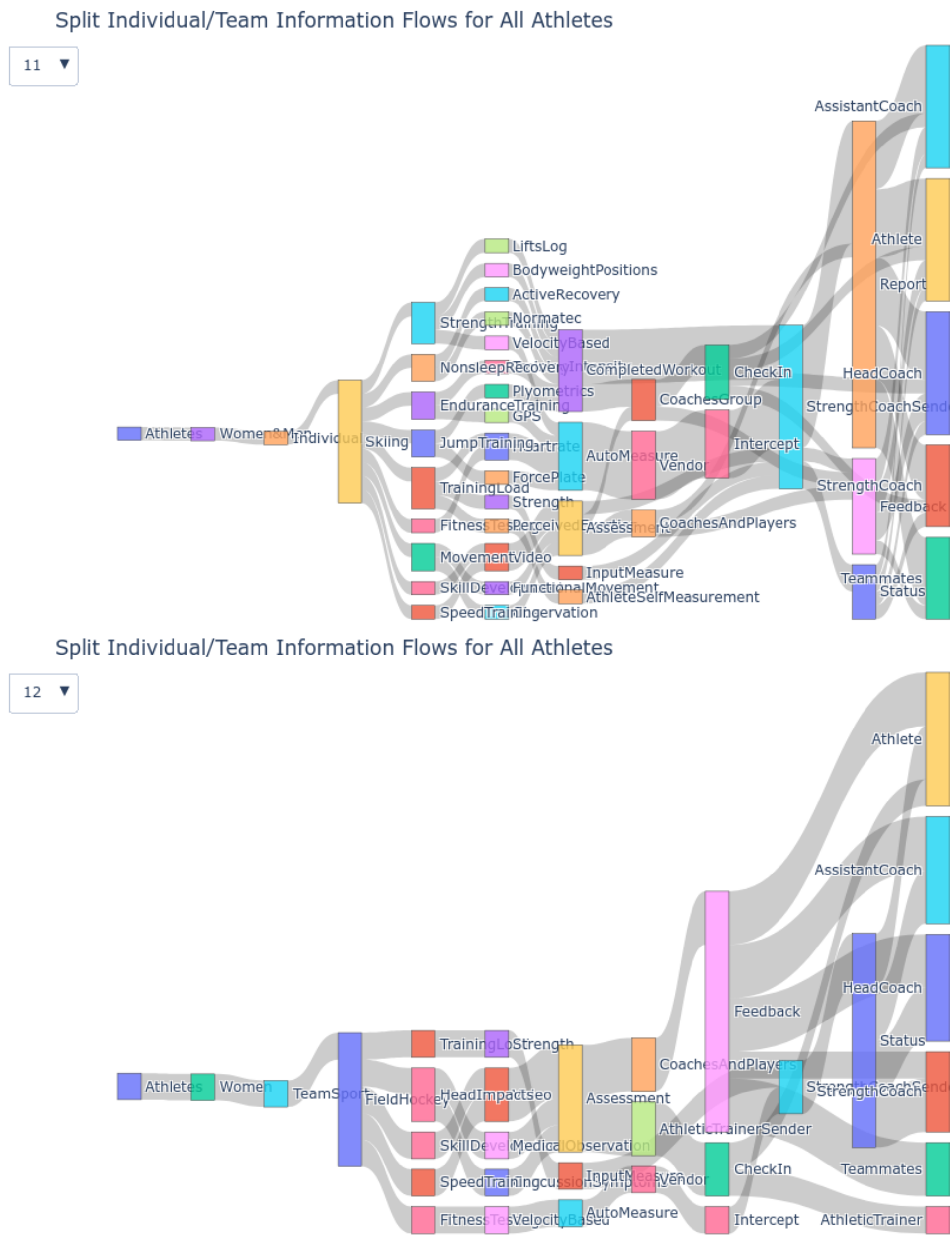


Fig. 10. Sankey Diagrams for Directed Graph Contextual Integrity Information Flows

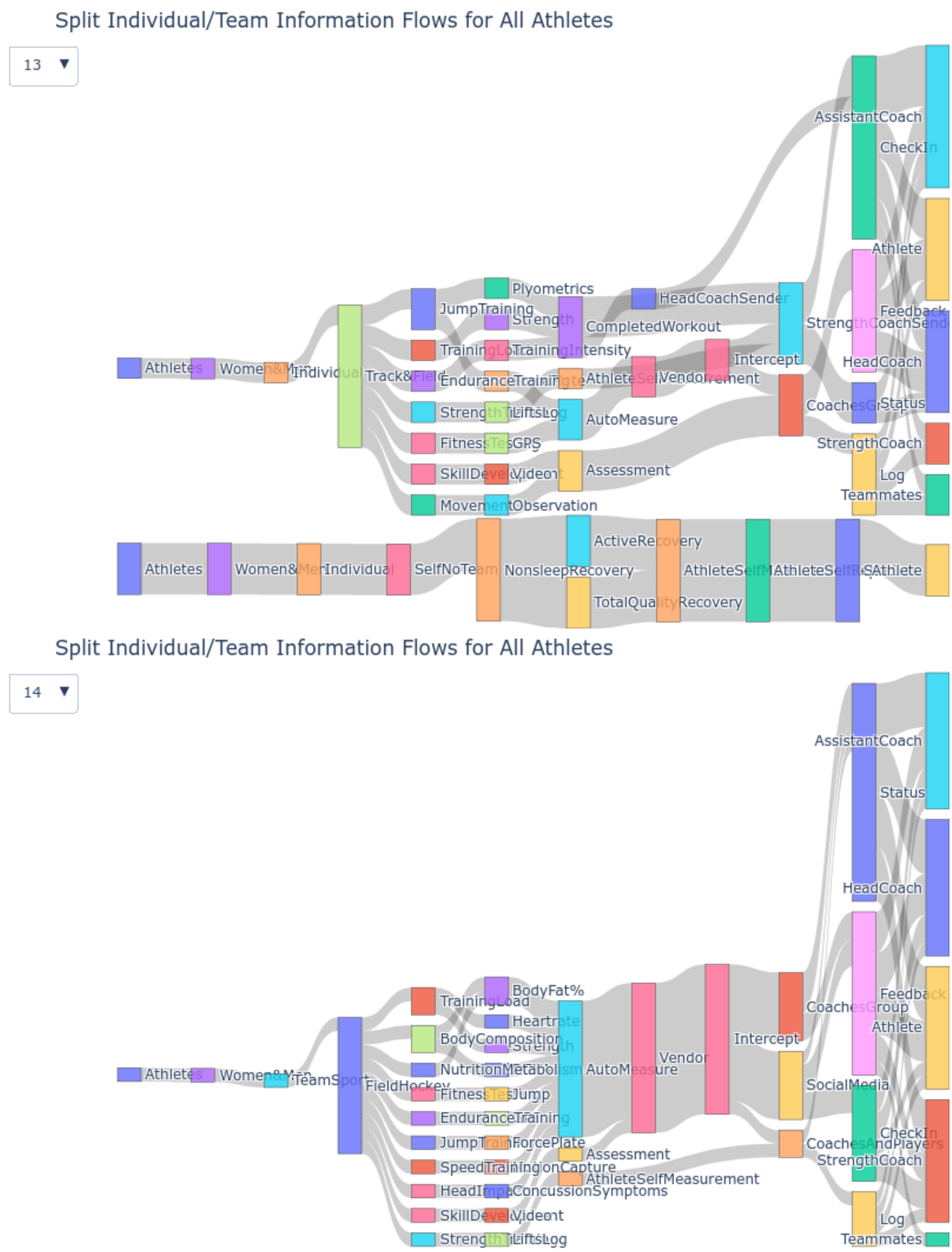


Fig. 11. Sankey Diagrams for Directed Graph Contextual Integrity Information Flows

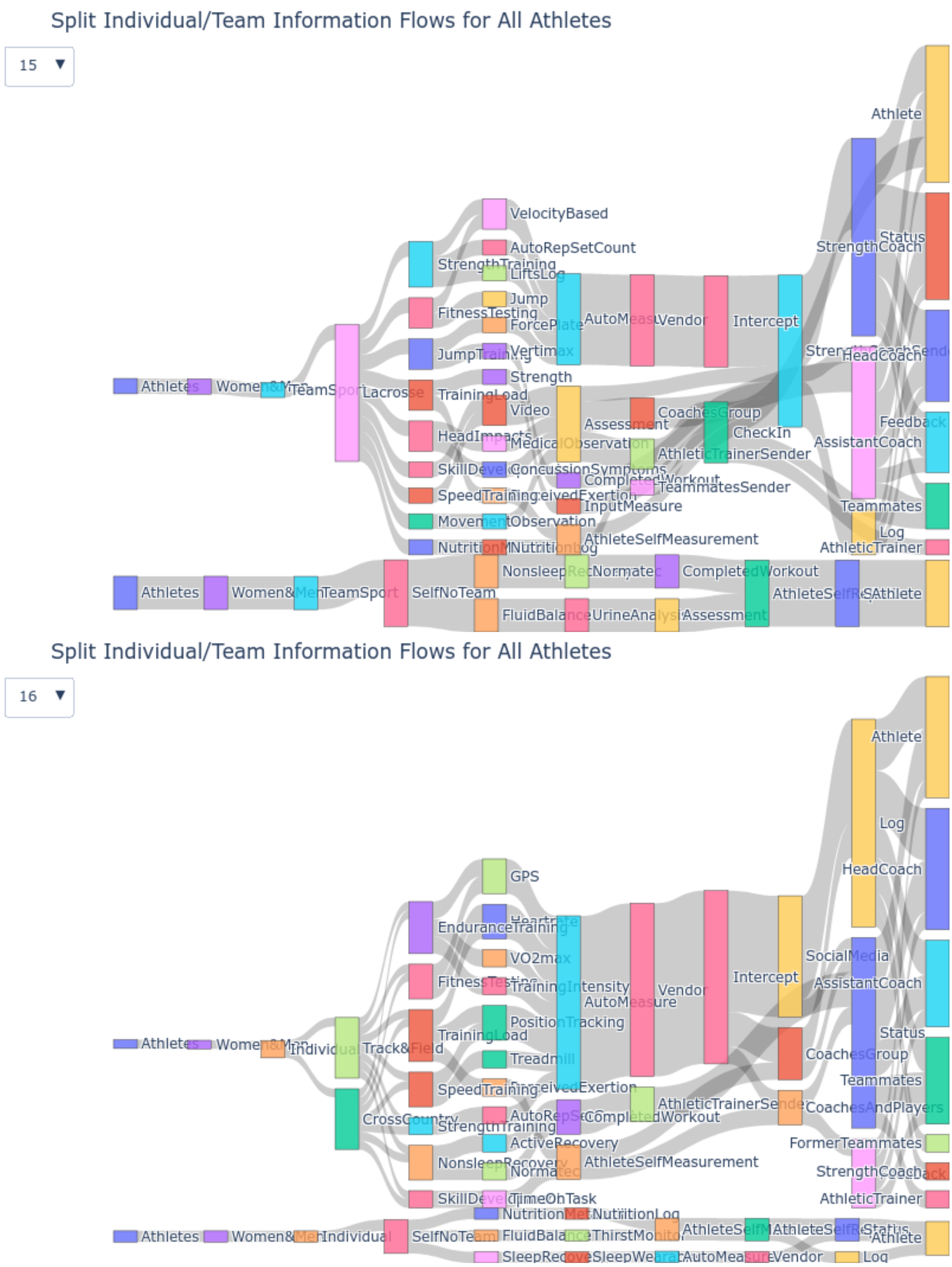


Fig. 12. Sankey Diagrams for Directed Graph Contextual Integrity Information Flows

D Appendix: Codebook

version	theme name	code id	name	definition	id	case	quote

1.0	Development	CHAL	Challenge	Examples of identifiable challenges to be overcome as part of development	10	U6 Nutrition	one of our is like you about your one else's bo pecially with lot of them s So we try an that topic.
					5	U4 Setbacks	I think for r tal mainly. S like really fo time and and goal just for make small g reach them.
		BIO	Biopsychosocial - Physical	refers to gains in strength, speed or another physical attribute	12	U3 Injury	if I haven't e performing a tition day, I' being so inj associated w yes, the spor for the game prepared, yo
					11	U1 Training Load	It's super ea You might ev might be feel it and. Over thing
		PSY	Biopsychosocial - Mental	Gains in skill, tactics, mental well-being, happiness	5	U4 Setbacks	I need to fo return to lo why I 'm pla I 'm doing this room an should be im bers versus physical
					10	U1 Training Load	feel like th between me readiness. An to do ment makes you fe ically better
		SOC	Biopsychosocial - Social	Gains in social connections and relationships	14	Foundation	I think I hav satisfaction i different gro feels like I'm a lot and I th feel better al

				5	U5 Leaderboard	I think it's with a team leaderboard, fine who yo not like mea right?
Collaboration	A-HC	Head Coach collab	Athlete-Head Coach collaboration	12	U4 Setbacks	there's lots hockey. Lots not just runn bowl like th so many opt things to do.
				15	Foundation	They kind o numbers an dence in yo vice versa. I the numbers you're not pl
	A-SC	Strength Coach collab	Athlete-Strength Coach collaboration	7	Foundation	one year the would rathe weight and c because I wa running and much.
				13	U5 Leaderboard	Yeah, not exp well and wh like that bla hmm. It's mo sheet of data for our times that's availa is available t
	A-AC	Assistant Coach collab	Athlete-Assistant Coach collaboration	2	U1 Training Load	The assistan don't make but he does. difference be what I was s the one I wa
				7	Foundation	I have two p tant coache them was a mate, so very where it's m sometimes.

	A-T	Athletic Trainer collab	Athlete-Athletic Trainer collaboration	16	U3 Injury	all the peopl and your the medical pers to like. Focus from getting kind of need
				15	U3 Injury	Michelle, sh ing up on pe an injury, progress she then after cleared, she's a week or tw
	A-TM	Teammates collab	Athlete-Teammates collaboration	1	U2 Sleep Recovery	I think if th all that, hey looking good can kind of a it helps whe roommates mates or we houses. So it
				10	U5 Leaderboard	when it com ing, it's we c eryone else' doesn't pre saying what you kind of c board in you
Information Flow	PUR	Purpose - General	speaks to Purpose as the central aspect of social context in an information flow	2	U5 Leaderboard	I think it se well. I thin someone els then starts trying to pr think it pron vironment.
				4	U1 Training Load	I think the p coaches and ganized and doing too mu themselves.
	BEN	Benefit	speaks to Benefit as identifier for Recipient in an information flow	10	U1 Training Load	the individu 'cause it give of their own coaches can mean, certai instances th better under and what's b

					13	U5 Leaderboard	I would say unless you're itive junkie fuel fuel thei
1.1	Gray Areas	CON	Conflict	describes an adversarial relationship between athlete and another potential collaborators	2	U2 Sleep Recovery	your coach k and like sees day and it's j fortable thin that you wer
					6	U6 Nutrition	because like that you war work on wit or like a nutr OK. But like coaches into get like turn area, especia with like his disorders.
					16	U4 Setbacks	What do yo area? Like ca kind of affect could be tech team's busin but also it's your person
		PC	Personal choice	examples and circumstances where athletes' personal choice should be respected; not yet an adversarial situation but could become one	2	U6 Nutrition	So much of revolves aro it's really ea to keep tabs tion is more choice. Ever dietary prefe etary goals.
					9	U1 Training Load	I think just co to the indi what they wa you know, I should alway dividual the
1.2	Set up for success	HAB	Habits	when asked about injury prevention and risk, responses usually focused on habitual behaviors that athletes felt gave them the best chance to avoid injury	10	U2 Sleep Recovery	The sleep ha sleep better. thing was lik of like going and waking my like beh sleep and lik doing to get

			9	U3 Injury	Injury risk is like how you your sleep ar opportunitie to mind, or like a higher ing a certain getting enou tion, nutritio
ROLE	Role Model	there are both positive and negative examples of people and behaviors for young athletes	8	U6 Nutrition	everyone yo to be like sup want to lift h room becaus to be bulky one's kind of ting stronger
			3	U2 Sleep Recovery	we have a c are like, yo initely have sleep pattern ing staff ar you should p melatonin c your sleep sc
			11	Foundation	I think tha many peopl sonality trai for instance less strong ity traits, but strengthened rounded by t
IS	Information Seeking	at a few points the participants asked for information or described how they actively sought information to answer a related personal question	8	U7 Women	I actually did study about gen levels c tears and I th per interestin how your hor like your lax and tendons cycle.
			14	U7 Women	I think it women und why they'r ways or like strong on o others.

		IMP	Improvement	athletic development has the expectation of improvement; recognizing improvement is the realization that development has occurred	13	U1 Training Load	I mean our load near ou our peak p odization pr we use those ments to ge how much w into our trai
					3	U1 Training Load	we use those beginning of of gauge like rest of the ample, we'll coach, our lif we're gonna squats at 60 ways refere original testi
		EXT	External Resources	some athletes have the means and the interests to incorporate external information sources (apps, wearable technology, or human experts) into their athletic development	10	U4 Setbacks	if we had a po would take m it so I could l sometimes I game or som look at the d and be like, really hard. give and tak
					8	U2 Sleep Recovery	I think that cool 'cause I can't change help my reco my sleep. So ing like a cr and my reco I know that I earlier.
					12	U6 Nutrition	If I know I h practice ahea eat. Thanks I know how
1.3	Differentiation	IM	Internal Motivation	when an athlete initiates behavior based on the personal choice and circumstances	9	U1 Training Load	I use it more I 'm just ma staying on to need
					12	U5 Leaderboard	You only con instead of w you're just th because you

ACC	Accountability	when a coach or team reinforces desireable behaviors that individual athletes may not follow through with on their own	11	U2 Sleep Recovery	I think our v most of our scious about sleep and th way to ensu good sleep.
			1	U2 Sleep Recovery	something al that you've a communit mates and you're all doi of the accou one to adopt
			14	U5 Leaderboard	I think eve one else's r starts talkin to prove th it promotes ment.
MUD	Mudita	feeling happiness for the success of others, as first expressed in Buddhist philosophy	5	U5 Leaderboard	The purpose is recognitio our teammat them.
			2	U5 Leaderboard	I guess the fa fit [from lea mean, all of because we a team to go.
EM	External Motivation	when an athlete initiates behavior based on the environment created by coaches and/or teammates	3	U1 Training Load	I believe tha like a bench yourself up a teammates h and like a be or like a trap going to try
			7	U4 Setbacks	I think that number in m mind could, y motivating.
			4	U5 Leaderboard	If I'm doing next to them it, beating th know I'm do

Presentation	PRES	Convenience	work required to privatize individual athlete information is not seen as worth the effort	1	U5 Leaderboard	there's a sh names and w time we got you can do that gets cre a mental lea
				8	U5 Leaderboard	we're all in t it's like one out the num like who's r OK and whe and stuff. W whiteboard see who's lea like, um, rea
				15	U5 Leaderboard	We did the t one can see one's kind other's score
	REC	Recognition	tangible reward to reinforce effort, contribution to team, and individual athlete development	5	U5 Leaderboard	if you do g to get sticke recognition. like having you're still for example, as many stic
				7	U5 Leaderboard	It's only ap light the fas say, and not like 3 fastest ate as oppos ing it all the ing it be see ing behind
	PUB	Public Information	individual sports like running and swimming have public information services that show athletes' race results	1	U5 Leaderboard	It's called li keeps track vidual perso they did and
				4	U5 Leaderboard	There's a s called Swim that's very a apparent. A swimmers d rankings.

				2	U5 Leaderboard	I guess the fa fit [from lea mean, all of because we a team to go.
	EM	External Motivation	when an athlete initiates behavior based on the environment created by coaches and/or teammates	3	U1 Training Load	I believe tha like a bench yourself up a teammates h and like a be or like a trap going to try
				7	U4 Setbacks	I think that number in m mind could, y motivating.
				4	U5 Leaderboard	If I'm doing next to them it, beating th know I'm do
Presentation	PRES	Convenience	work required to privatize individual athlete information is not seen as worth the effort	1	U5 Leaderboard	there's a sh names and w time we got you can do that gets cre a mental lea
				8	U5 Leaderboard	we're all in th it's like one out the numb like who's r OK and whe and stuff. W whiteboard see who's lea like, um, real
				15	U5 Leaderboard	We did the t one can see one's kind o other's score
	REC	Recognition	tangible reward to reinforce effort, contribution to team, and individual athlete development	5	U5 Leaderboard	if you do g to get sticke recognition. like having you're still p for example, as many stic

					7	U5 Leaderboard	It's only ap light the fas say, and not like 3 fastest ate as oppose ing it all the ing it be see ing behind
		PUB	Public Information	individual sports like running and swimming have public information services that show athletes' race results	1	U5 Leaderboard	It's called lik keeps track vidual perso they did and
					4	U5 Leaderboard	There's a sv called Swim that's very a apparent. An swimmers do rankings.
1.4	Role and Responsibility	DMR	decision-maker role	appropriate data-sharing for coach-collaborator's decison-making (not necessarily mutually beneficial)	6	U4 Setbacks	I would say could be ap shape their cisions or lil they believe where they r performing l
					8	U1 Training Load	I don't really data goes. Is that only go but like if th could proba where
		PLR	planning role	appropriate data sharing for coach-collaborator's planning (mutually beneficial)	7	Foundation	they kinda, at what we l training and where they'd from there ar where they'c
					11	U1 Training Load	our strength our strength training bloc our training duce the loa ery weeks an load and inte intensity blo ume week.

	ESR	expert support role	appropriate personal data-sharing exclusively with expert in support role (trainer, nutritionist, counselor)	4	U2 Sleep Recovery	my athletic she would ing my sleep would be abl like, this is re recovery, the need to sleep
				8	U4 Setbacks	Say you nee one day, 'cau in a great he your injury, with that.
				10	U6 Nutrition	I think of a r as similar as gist, kind of a one-on-one
	EXC	exceptions		2	U2 Sleep Recovery	I don't feel other people data. I think you have to
				3	U2 Sleep Recovery	at the end you're, you k going to ben directly from I got 8 hours
				12	U7 Women	I usually don information trainers beca be able to u mean by und
Relationship Dynamics	PROG	progression	athlete opionion reflects input from coach-collaborator re: athlete development progress	8	Foundation	I love doing to talk and of like whe progress is a like I share w be doing bet
				6	Foundation	It's kind of l both like fun ability and b they're here support you
				2	U1 Training Load	If their traini and stuff is we get faste good for the

ADPT	adaption	athlete recognizes need for changes based on data, seeks/accepts guidance from coach-collaborator re: adapting	3	U1 Training Load	To gauge ou out where ou ing and wh might use so It's a long se
			8	U1 Training Load	say we had practice and that or like a that in the lif conditioning
CON	consensus	athletes' privacy opinion reflects consensus among coaches and/or teammates	14	U1 Training Load	I think if tha sion and eve forward wit buy into tha
			12	U2 Sleep Recovery	We could be everyone an each one of coaching sta their own.
MAT	maturity	younger athletes gain confidence as they grow older, changing the dynamic in coach-collaborators	15	Foundation	at first it's l comfortable freshman ye dated. But I t our coaches
			10	U4 Setbacks	the next sum I used my ma edge I gained ciently and th ing.

Acknowledgments

To Robert, for the bagels and explaining CMYK and color spaces.

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Received May 2026; revised TBD; accepted TBD